UCD CHEM 4A LAB: GENERAL CHEMISTRY FOR MAJORS

Chem Annex Dispensary
University of California, Davis

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Chem 4A Lab: General Chemistry for Majors

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CHAPTER OVERVIEW

Chem 4A: Laboratory Manual

Preface and Acknowledgments

Preface

This manual contains instructions for the laboratory experiments that you will perform in Chemistry 004A. The experiments are designed to teach you a variety of laboratory and data analysis techniques. These instructions are meant to guide you in doing the experiments and interpreting data. You must read and understand the assigned experiment, and any relevant sections of the lecture text, before coming to the laboratory. You cannot perform the experiments safely and correctly nor make effective use of your laboratory time without such advance preparation.

Acknowledgments

This manual is the culmination of the efforts of many individuals. Many faculty members have provided ideas for the creation of these laboratories and have made numerous suggestions regarding their implementation. Stockroom Dispensary Supervisors, past and present, have had a role in helping to develop these experiments and, in particular, helping to ensure that the experiments are tailored to our laboratories at UC Davis. Safety TAs, past and present, have edited this manual to ensure that the experimental procedures are clear and current. In addition, many undergraduates have been involved in the development of these experiments and the completion of this lab manual as part of their undergraduate research projects.

Safety Rules for the Teaching Laboratories

Safety Rules

The following rules are designed for your safety and it is your responsibility to follow these rules. The Laboratory Instructor (LI = TA, Laboratory Supervisor, and/or Course Instructor) must enforce these rules, approved by Department of Chemistry Staff, Faculty, and Department Safety Manager. The LI must also enforce all experiment-specific safety rules pertaining to laboratory work, including rules not explicitly mentioned here. Violation of these rules will result in expulsion from the laboratory.

1. No one is permitted in the lab without the supervision of an LI. No lab work will be done without supervision. Perform only authorized experiments, and only in the manner instructed. Do not alter experimental procedures, except as instructed.
2. Specific permission from your LI is required before working in any lab section other than the one assigned to you. Only lab rooms where the same lab course is operating may be used for this purpose.
3. If you have a special health condition (asthma, pregnancy, etc.) or any personal health concerns, consult your medical professional before taking chemistry lab. If you require accommodations such as wheelchair access, service animals, or other, consult with the Department Safety Manager.
4. Students must come to the lab with appropriate lab attire and personal protective equipment (PPE). Without either, you will not be allowed to enter or make-up the lab, and your course grade will be affected. If you cannot wear the required PPE or lab attire, you must meet with the Department Safety Manager before you are allowed to enter the lab.
5. The appropriate PPE requirements include the following:
    1. Properly-fitted white 100% cotton lab coats. Lab coat sleeves must cover your wrist and should cover your thighs when seated. Disposable, flimsy, short, or dyed lab coats are not allowed.
    2. Approved safety goggles must be worn at all times. Safety goggles may not be modified in any manner. Visorgog, safety glasses, swimming goggles, or similar eyewear are not acceptable in the lab.
6. Lab attire must be made of durable material to protect from incidental chemical splash. Appropriate lab attire includes:
    1. Clothing that completely covers the legs - including the skin between the top of the shoe and the bottom of the pant leg - must be worn at all times in the lab. Leg coverings must extend to the shoes. Socks must be long enough to cover the ankles while seated. Tights, leggings, joggers, yoga pants, pants with holes, pants with slits, etc. are not suitable leg covering regardless of material. No-show socks are not allowed.
    2. Closed-toe, closed-heel shoes that completely cover the entire foot (including the top of the foot) must be worn at all times. Crocs, jellies, boat shoes, low-heel slippers, slides, slip-ons, etc. are not allowed. Socks are not suitable replacements for shoes or leg coverings.
    3. Because inadequate protection often leads to injury, it is recommended to ensure all skin below the neck is covered.
    4. Avoid wearing expensive clothing or jewelry to the lab as it may get damaged.
7. All students must adhere to safe lab practices, including the following:
    1. Horseplay and carelessness are not permitted and are cause for expulsion from the lab.
    2. You must be fully attentive and aware of your surroundings, and remove distractions as directed by your LI.
    3. Skateboards, rollerblades, motorized scooters, bicycles, and other transportation devices must be stored outside of the lab. Personal electronics are only permitted when needed for the lab. Cell phones or other personal electronic media can be damaged or contaminated in the lab. Use of such devices is at the student's own risk.
    4. Tie back long hair, loose hair, or hair that blocks your vision. Pin back loose headwear.
    5. Absolutely no food or drinks are to be stored or consumed in the lab. Contact lenses and cosmetics (hand lotion, lip balm, etc.) are not to be applied, and medications are not to be consumed while in the lab.
    6. Lab doors must remain closed except when individuals are actively entering or exiting the lab.
    7. The student must have at least one ungloved hand when outside the lab. Gloves are presumed to be contaminated and must not come into contact with anything outside the lab except chemical containers. Only use the ungloved hand to open doors, hold on to stair rails, or push elevator buttons.
8. Follow all instructions on safe chemical handling and safe equipment handling, including the following:
    1. Mouth suction must never be used to fill pipets.
    2. All activities in which toxic gases or vapors are used or produced must be carried out in the fume hood.
    3. Containers of chemicals may not be taken out of the lab except to the dispensary for refill/replacement or to exchange full waste jugs for empty ones. All containers must be closed with the appropriate cap before you take them into the hallway to

the dispensary. Always use a bottle carrier when transporting chemicals and waste.
4. Put hazardous waste into appropriate waste container(s) provided in your lab. Do not overfill waste containers.
5. Keep your working area clean – immediately clean up all spills or broken glassware. Dispose sharps in the appropriate container. Do not dispose pipette tips in regular trash. Clean off your lab workbench before leaving the lab.

9. Learn how to respond to emergencies and report incidents promptly to the LI:
   1. Learn the location and how to operate the nearest eyewash fountain, safety shower, fire extinguisher, and fire alarm box. Basic first aid for any chemical splash is to wash the affected area for at least 15 minutes and seek medical attention. Use the emergency shower if appropriate, removing contaminated clothing for thorough washing. If the safety shower or eyewash is activated, the exposed person should be accompanied to the Student Health Center for further evaluation.
   2. All incidents, near misses, injuries, explosions, or fires must be reported at once to the LI. In case of serious injury or fire (even if the fire is out), the LI must call 911. The student must always be accompanied to the Student Health Center.

Students must sign the Safety Acknowledgement sheet before working in the lab. If you have any questions, please ask your LI before starting any lab work.

Safety Data Sheets

Each lab in this manual lists some of the hazards you can expect to see in the lab. As part of your preparation for the lab, you are expected to look up the SDSs for any chemicals used in the lab. These sheets contain important information on the hazards, first-aid measures, what to do if there is a spill, and the correct PPE to wear for that compound. Your TA may ask you about the hazards associated with the compounds you are using in the lab, and you are expected to be prepared to discuss them. All SDS are available online at https://safetyservices.ucdavis.edu/article/safety-data-sheets or https://ehs.ucop.edu/sds/#/.

Laboratory Notebook and Reports

Laboratory Notebook

Your laboratory data must be recorded directly into a bound, quadrille laboratory notebook that has duplicate ("carbon-copy") pages. **Reserve the first few pages for a table of contents, which should be kept up-to-date.** Entries should be legible and clear and made when an operation is performed or when the data is generated. These entries should be made in your lab notebook and not on a loose piece of paper to be copied later. All entries must be in black or blue ink in the English language. Each page should have the day's date at the top. If entries are made at a later date, a new page should be used. Always include units of measurement when recording data. Do not include any data that is not relevant to the lab, including pictures, drawings, or editorials. You should not use your lab notebook to take notes. Errors should be crossed out with a single line, not made illegible, just in case the "errors" later turn out to be important, and the correction should be written near (above, below, or to the side if possible) the original entry. The original information should still be able to be read. In industrial laboratories, it is required that these corrections have a written explanation of why the correction was needed (typically as a footnote). It is at the discretion of your TA whether this will be required. Most data should be handwritten, but it is permissible to tape a page into your notebook if the 4 corners are marked so that it will be obvious if the page is removed. You should also initial and date across one side of the page so that the partial initials and date are visible on both the taped page and the notebook page.

All information regarding an experiment should be recorded in your notebook. Prior to the lab, you should carefully read through the lab manual and prepare your notebook with the following parts.

1. <u>Title of Experiment:</u> The experiment title this pre-lab is for.
2. <u>Brief Description of Tasks:</u> 3-4 sentences summarizing the experiment. You may include the main objective of the experiment, concepts demonstrated, and a breakdown of major steps to complete main objective.
3. <u>Outline:</u> Experiment procedure steps in your own words. Make it detailed enough that you more or less able to perform the experiment with just these steps, but no so detailed as rewriting the lab manual.
4. <u>Sample Calculations (if necessary):</u> Some experiments will require calculations. The equations and calculations of hypothetical values should be shown here.
5. <u>Table of Reagents:</u> Reagents' chemical name or formula, physical state and amount that will be used in the experiment. These are found in the manual for each experiment. You can then google all other information missing. *Note: A reagent may be used multiple times. You can list it once and sum the amount. It also may be used multiple times but in different physical states. You can use the same row, but clearly state the amount used for each physical state.*

Your TA will check your notebook prior to entry into the lab. Any student who has not completed this notebook preparation is unprepared to safely perform the assigned experiment and will have to leave the laboratory until they have finished preparing.

At the end of the laboratory period, your TA must initial each page of your notebook that was used during the lab to confirm that the work was completed by you during the lab.

Lab Reports

Your lab report should include the following parts:

1. Title of the experiment - This should capture the essence of the work being reported. Be specific. An article's title is important for indexing and retrieval purposes and should include important keywords about the experiment performed. The first author (you) has an asterisk next to your name; Partners should be listed as co-authors. The name of the institution and the date go below.
2. Purpose of the experiment - What is your goal for the lab?
3. Outline of the procedure - A paragraph or two summarizing the important parts from the lab manual. This should not be word for word but rather enough information that someone else could replicate your results. This includes the actual measurements performed in the lab (if the manual says to weigh 0.1 g and you weighed 0.1103 g, you should report here that you measured 0.1103 g).
4. Data and observations section - It is advisable to make data tables in your lab notebook before coming to the lab, as they will make it simpler when you enter the data you collect.
5. Calculations section.

6. Discussion - Analyze your data in the context of the purpose of the lab. Quantify how well your computed/measured values compare to literature (this includes statistical analysis such as percent error and standard deviation). Include discussions on possible sources of error in the execution of the experiment and how they might have affected the outcome. How might they be avoided in the future? This section should include answers to the lab manual questions - Show your work and provide answers.
7. Figures - These can be included in the discussion or added at the end as an appendix and referenced in the report. If you include a figure, it should be referenced in the text.
8. Conclusions - State the main findings of the experiment, and relate them to the purpose in the introduction. Concisely summarize your work.

Your TA may have specific instructions for the reports of each individual lab, but the format above should generally be followed. Turn in your report and the carbon copies of your lab notebook to your TA one week following the scheduled completion of that experiment.

You will perform your experiments with lab partners, and it is encouraged that you should discuss your data, analysis, and interpretation with your partner, with other students in the class, or with your TA. However, all work performed and reported should be your own. Plagiarism of any kind will not be tolerated and will be reported to Student Judicial Affairs.

Statistical Treatment of Data

Introduction

In physical and analytical chemistry, you are often left with large sets of data you will need to transform into useful information. This will require some manipulation of data and some calculations. Understanding your data begins with understanding that a histogram of an infinitely large number of good measurements typically follows a Gaussian (Normal) distribution. The equation for this distribution is given by:

$$f(x_i) = \frac{1}{\sqrt{x\pi\sigma}} e^{-\frac{(x_i-\bar{x})^2}{2\sigma^2}} \tag{1}$$

where σ is the standard deviation and $\bar{x}$ is the average of your data, both defined below. Important features from this distribution are that the function is centered at the true average of the measurements, and the width of the distribution is related to the standard deviation, or the clustering of the data.

The reason this matter is a finite set of data will also follow this distribution, but there is uncertainty in where exactly the true average lies, and what the true deviation from it is, without doing an infinite number of measurements. When you report data, you will want to report it in these terms, taking the uncertainty in the true value into account. Below are some procedures for averaging data, finding the standard deviation, finding the relative deviation, finding the confidence interval, and analyzing poor data.

Averaging Data

The arithmetic mean, often called the average, is the sum of the measurements divided by the total number of measurements:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{2}$$

where $\bar{x}$ is the mean of the measurement of x, x_i is the ith measurement of x, and n is the total number of measurements being averaged. When you take repeat measurements of your data, you should calculate the average for your results for your report.

Standard Deviation

The standard deviation, σ, measures how close values are clustered about the mean. The smaller your standard deviation, the higher the precision of your measurements.

The standard deviation for small samples is defined by:

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(\bar{x}-x_i)^2} \tag{3}$$

The square of the standard deviation is called the variance.

Relative Standard Deviation

You sometimes will want to convert the standard deviation into a relative standard deviation to compare how precise it is compared to other measurements of different magnitudes. This is calculated by:

$$RSD = 100 \times \frac{\sigma}{\bar{x}} \tag{4}$$

Confidence Interval

With a finite number of data points, it is not possible to find the true mean or the true standard deviation, as a small sample size will not cover the entirety of the normal distribution of data. What we calculated in the earlier sections is the sample mean and the sample standard deviation. The confidence interval is an expression that states that the true mean is likely to lie within a certain distance from the measured and reported mean, $\bar{x}$. The confidence interval is given by:

$$\mu = \bar{x} \pm \frac{t\sigma}{\sqrt{n}} \tag{5}$$

where μ is the interval, and t is the students t, which can be looked up in the student's t table (available in many textbooks or online). The value of t for a 95% confidence interval can be found in Table 1:

Table 1: Student t table for a 95% confidence interval.

degrees of freedom $(n-1)$	95% Confidence Interval t Value
1	12.706
2	4.303
3	3.182
4	2.776
5	2.571
6	2.447
7	2.365
8	2.306
9	2.262
10	2.228
15	2.131
20	2.086
25	2.060
30	2.042
40	2.021
60	2.000
120	1.980
∞	1.960

Discarding Data

You should always be cautious of discarding data. Just because a data point is an outlier does not mean it is wrong; that may turn out to be your most accurate measurement. Two methods are generally accepted when deciding to reject data. The first is rejecting data that you know to be low quality due to the procedure of measuring it. You can do this by adding notes in your lab notebook, such as any issue in the lab, extra solvent added, a little bit of spilled powder, or instrument malfunctions, leading to problematic data collected. If you know you did something wrong, you can discard that data as bad data.

The other method for rejecting a single outlier (you can only perform this procedure once on a dataset) is called the q-test. For $3 \leq n \leq 10$, where n is the number of measurements of the same quantity, calculate:

$$Q = \frac{|\text{suspect value} - \text{value closest to it}|}{\text{highest value} - \text{lowest value}} \tag{6}$$

Compare the value of Q with Q_c from Table 2. If $Q > Q_c$, you can reject the suspect value.

Table 2: Critical Q values for rejection of a discordant value at a 90 % confidence level

n	Q
3	0.94
4	0.76
5	0.64
6	0.56
7	0.51
8	0.47
9	0.44
10	0.41

Significant Figures

When reporting data, you should only report the number of figures that are "significant." The number of digits used in a number denotes the highest precision that the number can be determined. For a balance that is accurate out to 0.0001, the precision of its measurements extends to the ten-thousandth place. There are many rules governing how significant figures can be found and propagated through a calculation, but for this class, we will use a simplified method in your reports.

To decide how many decimals to include in your reported measurements:

1. Calculate the standard deviation (or confidence interval for a finite number of data points).
2. Truncate the standard deviation (or CI) to either 1 or 2 significant figures. If truncating the sig figs to 1 significant figure would result in that digit only being a 1 or a 2, you will want to leave it at 2 significant figures.
3. Report your mean as the same number of decimal places as your standard deviation.

So, for example, if your mean is 10.12345 ppm, and your standard deviation is 0.032 ppm, you will report your measurement as 10.12 $\pm$ 0.03 ppm (don't forget units). Or, if your mean is 10.3452 μV, and your CI is 0.0194 μV, you will report your measurement as 10.345 $\pm$ 0.019 μV.

1: Calibration of Volumetric Glassware (Experiment)

Hazard Overview

Chemical Hazards

N/A

Mechanical Hazards

N/A

PPE

1. 100% cotton lab coat
2. Splash goggles
3. Nitrile gloves

Dispensary Provided Items

- 125 mL Erlenmeyer flasks
- Flask stoppers
- 50 mL Burette
- 25 mL Pipettes
- Pipette bulb
- Thermometer
- 300 mL beaker
- Wash bottle

Introduction

You will use a variety of volumetric glassware and other equipment in your laboratory experiments. Whenever possible, you should ensure that the equipment is properly calibrated so that systematic errors do not result from using the equipment to make measurements. In this experiment, you will learn how to calibrate both your pipette and your burette. The process demonstrates the importance of careful technique and the types of corrections that are needed in accurate laboratory measurements. You also will compare your pipette result with that of other students to investigate the scatter associated with nominal 25 mL pipettes. You will interpret this scatter in terms of the theory of random errors.

Calibration is quite simple. The pipette or burette is used to deliver a certain volume of water. The actual volume is determined by measuring the mass of the delivered water and calculating the volume from the known density. There are, however, several factors that must be considered for accurate results. First, the mass of the water must be accurately determined. Although seemingly straightforward, given a good balance, careful technique is required, and corrections must be applied to compensate for buoyancy effects. Second, since the density of water depends on temperature, its temperature must be known to calculate its volume. Thirdly, the glass itself expands with increasing temperature. Hence, the volume delivered by the pipette or burette will change with temperature.

Read and understand the operation of an electronic balance before you begin this experiment.

Operation of Lab Equipment

Analytical Balance

The operation instructions can be found below. **Do not attempt to calibrate the balances used in this course. This has already been done for you and is checked regularly by dispensary personnel.**

1. Check that the balance is level before using it since the calibration depends on the balance being level. Notify your TA if the balance is not level.
2. Ensure that the balance and the area around it are clean. Notify your TA if the balance is not clean, and they will instruct you to clean it.
3. The doors on either side of the balance and the top all slide backward to give access to the balance. While weighing, these doors should all be closed.
4. Before weighing your sample, ensure that the weight displayed is 0.0000 g. The 0 or tare buttons on the bottom of the balance will zero the reading. Be sure to tare your weighing vessel before weighing your sample.
5. There will be a small circle on the display if the weight is changing. Once the weight has stabilized, this circle will disappear, and the mass can be recorded.
6. Do not bump or spill chemicals on the balance. If an accident occurs, let your TA know immediately.

Experimental Procedure

Careful technique is required to accurately use a pipette or a burette. Your TA will demonstrate the correct technique and explain precautions you should take using them. You should practice using a pipette to deliver exactly 25.00 mL of deionized water several times before you collect the actual calibration data. The largest sources of error in this experiment involve the improper use of the pipette and incorrect reading of the burette.

Draw about 500 mL of deionized water into a clean beaker from the Nalgene container by the sink. Use this equilibrated water to collect all calibration data.

<u>Before beginning lab, your TA will instruct you in the use of the pipette, the burette, and the analytical balances in the lab.</u>

Pipette

Calibrate your 25.00 mL pipette. Confirm that your pipette is clean by filling it above the mark with deionized water and then letting it drain.

Some important tips when using a pipette:

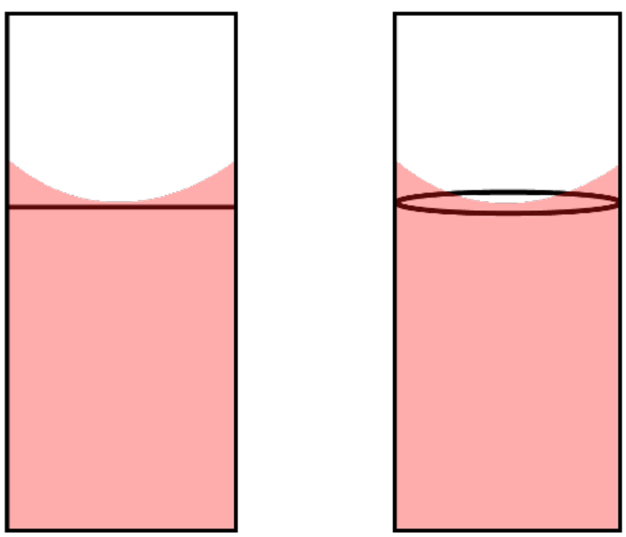


Figure 1.1: Placement of the meniscus. The left view shows what it looks like from the side, while the right view shows what it looks like isometrically.

- When you fill it with the bulb, fill well past the mark but not into the bulb. This will give you the most time to make a seal with your finger after removing the bulb.
- After filling the pipette with the bulb, quickly remove the bulb and stop the flow of water with your finger (index finger or thumb work best for most people). Use either a twisting or a rocking motion to let a small amount of water drain until the meniscus barely touches the line. See Figure 1.1.
- Don't push the bulb onto the pipette too far. It only needs to make a seal, and this doesn't require much overlap. This will facilitate an easier transfer using your finger.

- The heat from your hands is enough to change the temperature of the water through the walls of the pipette. Therefore, **minimize contact with the pipette below the water line.**

The steps for your pipette calibration are as follows.

1. Choose a clean 125 mL Erlenmeyer flask. Fit the flask with a plastic stopper, and weigh this assembly to the nearest 0.1 mg (or 0.0001 g). **The scales are sensitive enough to measure the difference in weight due to your fingerprints, so use nitrile gloves when handling the flask.** Use the same balance throughout the experiment. Avoid getting the outside of the flask or the stopper wet during the calibration procedure, as evaporation before weighing the water can affect your results.
2. Record the temperature of the beaker of DI water that has been sitting at room temperature. You will use this value and the density in Table 1.1 to convert your mass of water to the volume of water.
3. Use your clean pipette to deliver exactly 25.00 mL of the equilibrated water into the 125 mL Erlenmeyer flask. Touch the pipette tip to the side of the flask below the neck during delivery. **Remember that the pipette is calibrated to deliver so that the last drop remaining in the tip of the pipette after delivery should not be blown out.**
4. Stopper the flask after delivery to prevent evaporation of the water before its mass is determined. Weigh the stoppered flask containing the 25.00 mL of water to the nearest 0.1 mg.
5. Repeat the calibration at least three times. Each lab partner should perform their own calibration.

note: Four 25.00 mL deliveries can be made into the flask before the total mass exceeds the range of the balance.

Pipette Calibration Calculation

The density of water as a function of temperature is given in the second column of Table 1.1. Use the density for your temperature to calculate the volume of water delivered by your pipette in each determination, the mean volume, the standard deviation, and the 50% and 95% confidence limits for the mean. If one of your determinations appears inconsistent with the others, use the Q test to see if it can be discarded and consider repeating the calibration an additional time. Interpolate between the values in Table 1.1 if required.

Table 1.1: Density and Corrected Volume of Water

Temperature °C	Density g/mL
18.0	0.998599
19.0	0.998408
20.0	0.998207
21.0	0.997996
22.0	0.997774
23.0	0.997542
24.0	0.997300
25.0	0.997048
26.0	0.996787

Burette

The next procedure involves the calibration of your burette. Confirm that the burette has no clogs by letting it drain. There should be no water drops left on the inside walls.

Some important tips when using a burette:

- The marks on a burette start at 0 at the top and go down to 50 at the bottom. The value is read as the volume delivered by the burette. The volume delivered is found by taking the initial burette volume and subtracting it from the final burette volume.
- While there are marks to the nearest 0.1 mL on the burette, it is possible to interpolate between the marks to the nearest 0.02 mL delivered. Ask your TA for help with this. Burette volumes are recorded to two decimal places.
- Make sure there are no bubbles in the tip of your burette, as this can cause inaccuracies in the volume delivered if they come out during delivery.

- When you record the position of the meniscus, be sure to have your eye parallel to it to avoid parallax errors.
- For your safety, do not climb on the counter or stool when filling up the burette, and do not pour towards your face.

Burette Delivery Calibration

You will repeat the following procedure for 4 volumes of approximately 15, 25, 35, and 45 mL.

1. Fill the burette past the zero mark with equilibrated deionized water. Open the stopcock and withdraw sufficient water from the burette to remove any air bubbles from the tip. Adjust the liquid level in the burette at or slightly below the zero mark. Wait about 15 seconds for the water inside the burette to drain down the side walls. Record the level of the meniscus, estimating its position to the nearest 0.02 mL. Touch the tip of the burette to the wall of a beaker to remove any adhering drop of water.
2. Empty your weighing flask, wipe the inside neck dry, and reweigh the empty flask with its stopper to the nearest 0.1 mg.
3. Recheck the reading of your burette. There should be no noticeable change from your earlier reading (<0.2 mL difference). If a change has occurred, ask your TA for instructions before proceeding.
4. Use the burette to deliver the approximate amount of water into your weighed flask (see Table 1.2). Remove any pendent drop from the tip of the burette by touching it to the inside wall of the flask below the neck. Stopper your flask.
5. Allow time for water to drain down the walls of the burette, then record the level of the meniscus to the nearest 0.02 mL.
6. Weigh the stoppered flask with the delivered water to the nearest 0.1 mg.
7. Refill your burette to the zero mark or just below.
8. Repeat for each measurement.

Table 1.2: Burette Volume Calibration Data

Target Volume	15 mL	25 mL	35 mL	45 mL
Volume reading from the burette (mL)				
Mass of water delivered by the burette (g)				

Burette Uncalibrated Region Measurement

The final procedure is to determine the volume of the uncalibrated portion of your burette between the 50 mL mark and the stopcock. This is represented by the shaded region in Figure 1.2. Table 1.3 shows the data you will record and calculate from this procedure.

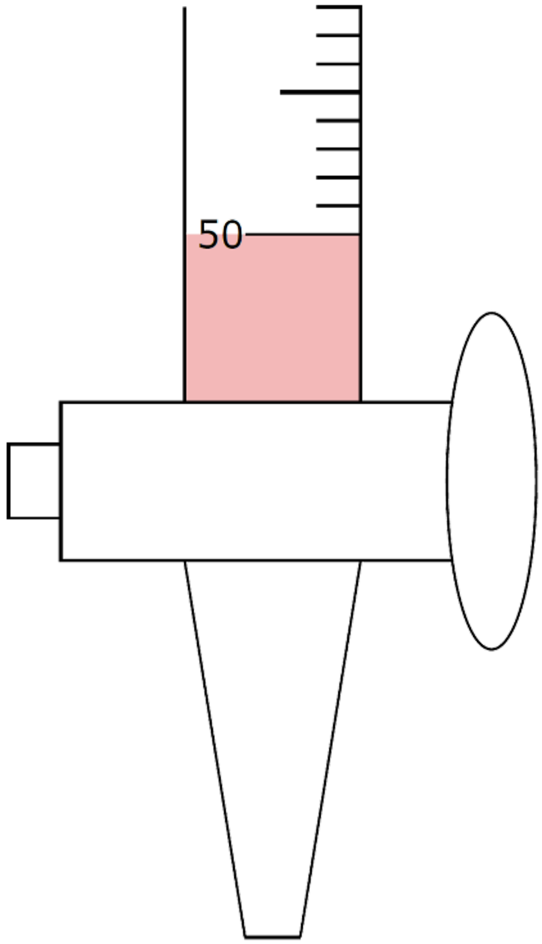


Figure 1.2: Uncalibrated region of a burette. The shaded region represents the region you will be measuring in this part of the lab.

1. First, empty your weighing flask, wipe the inside neck dry, and reweigh the empty flask with its stopper to the nearest 0.1 mg.
2. Fill the burette with about 10 mL of water and drain the water into a beaker until the level falls about 1 mL above the 50 mL mark. Remove any pendent drop, wait for drainage, and record the volume above the 50 mL mark to the nearest 0.02 mL.
3. Drain the water from the burette into the weighed flask, stopping when the water level is at the top of the stopcock.
4. Remove any pendent drop by touching the tip to the side of the flask, stopper the flask, and then **weigh** the stoppered flask with the delivered water to the nearest 0.1 mg.
5. Repeat the procedure three more times.

Table 1.3: Burette Uncalibrated Region Volume Data

	Run 1	**Run 2**	**Run 3**	**Run 4**
1: **Volume of water above the 50 mL mark on the burette (50.00 mL - initial volume in burette)**				
2: **Total Mass of water delivered by the burette (g)**				

3: Use the density equation, the theoretical density at your recorded temp from Table 1.1, and mass recorded in Row 2 to find the **total volume of water delivered (mL)**				
4: **Actual volume of water in the uncalibrated region (Row 1 - Row 3) (mL)**				
5: **Average volume (from Row 4) for all four runs. (mL)**				
6: **Standard deviation in Row 5 (mL)**				
7: **50% confidence limits in Row 5 (mL)**				

Lab Report

See Laboratory Notebook and Reports for details on what is expected of you in a general lab report.

Include the following in your **lab report**:

- Plot (scatter plot) the calculated burette volume (from recorded mass and theoretical density) vs. the delivered volume (final - initial volume readings from burette).
- Create a histogram of calibrated pipette volumes for the class. You should receive a link to Google Sheets to input your results and get your classmates' results. Wait until you have at least 10 values from your classmates before making this chart. This should look Gaussian, if it doesn't you need to change the bin size.
- Average density of water from your pipette measurements. Compare this with the value of the density of water at the temperature recorded in lab from Table 1.1. Find percent error, and comment on your accuracy. Include sources of error and how to minimize them.

Also, include the answers to the following questions in your lab report.

1. The internal diameter of the neck of a 25 mL pipette around the calibration mark is 5.0 mm. Assume that the meniscus can be positioned to ±0.2 mm relative to the calibration mark. If the error in positioning of the meniscus is the only error that occurs in the calibration of a pipette, how closely should duplicate calibrations agree?
2. Using the values in the table below, if the temperature has an uncertainty of ±1°C, what is the uncertainty in the burette's delivered volume for the 0-15 mL interval?
Initial Burette Reading	0.14 mL
Final Burette Reading	15.17 mL
Weight of the Flask	50.636 g
Weight of the Flask + Water	65.653
Temperature	22.54°C

3. An alternate way to determine the volume of the unmarked portion of the burette is to measure the diameter and height of the corresponding cylinder and simply calculate the volume of the cylinder. Assume that the diameter of the cylinder is 11.0 ± 0.3 mm and the height is 55.4 ± 0.4 mm. What is the volume of the region and its associated error in mL?

2: Charge and Mass of an Electron (Experiment)

Hazard Overview

Chemical Hazards

- Sulfuric acid - corrosive
- Acetone - flammable
- Copper sulfate - environmental hazard

Mechanical Hazards

- Electrolysis setup - electrical hazard

PPE

1. 100% cotton lab coat
2. Splash goggles
3. Nitrile gloves

Dispensary Provided Items

In the Fume Hood

- 1M sulfuric acid
- Acetone
- 1M copper sulfate in 0.5 sulfuric acid
- DI water
- Wash beakers

In the Lab

- The electroplating apparatus will be set up at your lab station
- Copper screens x2
- Tweezers

Introduction

In 1897, J. J. Thomson performed some pioneering experiments on cathode rays, which yielded the first measurement of a physical constant for the electron. He showed that cathode rays were composed of a stream of negatively charged particles, later called electrons, which were deflected by electric and magnetic fields. From his studies, he obtained the charge-to-mass ratio for the electron, which has a currently accepted value of $e/m = 1.7588196 \times 10^{11} C/kg$. Since Thomson's experiment could only give the ratio, a separate experiment was necessary to determine either e or m to have their individual values. Twelve years later, R. A. Millikan succeeded in measuring the charge of the electron with his famous oil drop experiment. The experiment you will perform will also yield a value for the charge of the electron. Given its charge, you can then also calculate the mass of the electron from the accepted value for e/m.

The technique that you will use to determine the electron's charge is based on the electrolysis of an aqueous solution of copper sulfate, $\mathrm{CuSO_4}$. This solution contains the ions $\mathrm{Cu^{2+}(aq)}$ and $\mathrm{SO_4^{2-}(aq)}$. Elemental copper, Cu, can be deposited from this solution onto an electrode, a process called electroplating, by supplying two electrons per $\mathrm{Cu^{2+}}$ ion. The electrons can be supplied by a battery or, as in the present experiment, by a power supply. By measuring the amount of Cu deposited and the total electric charge passed into the solution, the charge per electron can be calculated.

The amount of Cu deposited is determined by weighing the electrode before and after the plating. The total electrical charge, or the number of coulombs passed during the plating, depends on the current and the amount of time it is running. One coulomb is defined as the quantity of charge that crosses in one second a section of a conductor in which there is a constant current of one amp.

$$Q = I \times t \tag{2.1}$$

Where I is the current and is constant, Q is the total charge in coulombs, and t is the time during which the current is flowing. If the current is not constant, a measurement of the number of coulombs can be calculated in LabQuest by integrating the current over time. Either method is acceptable for this experiment.

A schematic of an apparatus for the electrolysis of copper is shown in Figure 2.1. Two copper-screen electrodes are partially immersed in an acidic electroplating solution containing $CuSO_4$ and connected to a power supply. The external power supply produces a constant current to flow through the cell. The current is carried in the external circuit (wires) by the movement of electrons and in the solution by the movement of ions. Note that the movement of positively charged cations in one direction in the solution is equivalent electrically to the movement of negatively charged electrons in the opposite direction in the wires.

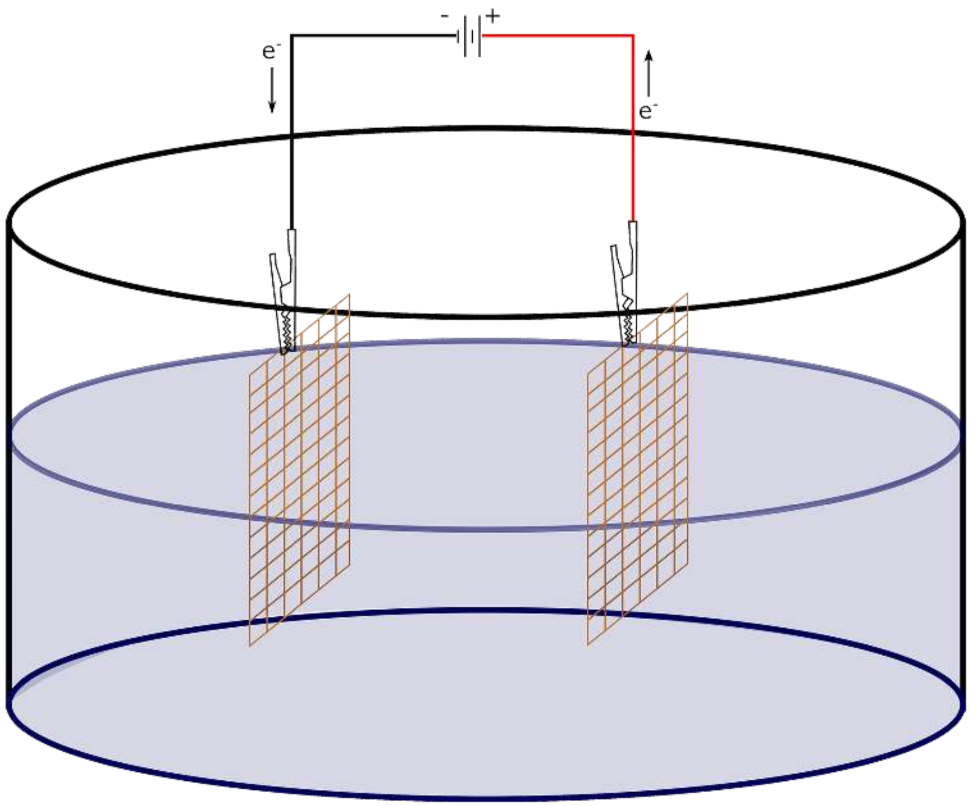


Figure 2.1: Schematic of Electrolysis Apparatus (CC BY-NC-SA;??????)

Copper cations in the solution migrate to the negative electrode, where they are reduced and deposited as elemental copper, according to the half-reaction:

$$Cu^{2+}(aq) + 2\,e^{-} \longrightarrow Cu(s) \tag{2.2}$$

The electrode at which the reduction occurs is called the cathode.1 the other electrode, the anode, copper from the electrode is oxidized and goes into solution according to the half-reaction:

$$Cu(s) \longrightarrow Cu^{2+}(aq) + 2\,e^{-} \tag{2.3}$$

Ideally, the anode loses mass, and the cathode gains an equal mass, as copper is transferred from the anode into the solution and from the solution to the cathode. However, complicating reactions occur at the anode. You should therefore report the mass gain at the cathode.

Although you will learn more about the requirements of the power supply in a later quarter when you study electrochemistry and thermodynamics, some qualitative comments can be made now. Sufficient voltage must be applied for current to flow through the cell (the electrodes and the solution), namely to overcome the electrical resistance of the cell and any energy barrier for the oxidation/reduction to occur. Once this voltage is exceeded, the current will rise linearly with the applied voltage. Since the current measures the flow of electrons in the connecting wires and of ions in the solution, the rate at which copper is deposited depends on the current. In your experiment, the current stays at a constant and high enough value for the deposition to occur in a reasonable time. This occurs since the concentrations of $CuSO_4$ and H_2SO_4 in the solution are large and constant and the electrodes at a fixed separation, so that the resistance of the cell is small and constant. Hence, so long as the power supply delivers a fixed voltage, the current also remains fixed.

Operation of Lab Equipment

For this lab, you will be using the Logger Pro software coupled with a current probe in line with the electrodes. This will allow you to monitor the current versus time for your
experiment.

Experimental Procedure

1. Begin by selecting 2 copper screen electrodes from the lab supplies and cleaning them with the following procedure in the fume hoods at the back of the lab:
    - Clean your 2 copper screens by using tweezers to place them in the dish containing 1M H_2SO_4 for 2-3 minutes.
    - Remove the screens with the tweezers, and dip it in the deionized water dish then swirl a few times.
    - Do a final rinse with deionized water under the faucet. Pat dry with paper towels.
    - Use the tweezers to swirl the screens in the acetone-containing beaker to remove excess water.
    - Once the copper screens have air-dried, they are ready to be weighed.
2. Weigh each of your electrodes to the nearest 0.1 mg (0.0001 g) using the analytical balances found at the front of the lab. Be sure to keep track of which electrode is which.
3. Fill a 250 mL beaker with about 100 mL of aqueous electroplating solution (it doesn't need to be exact).
4. Check your electroplating setup and verify that your lab quest is connected to the PC through the USB.
5. Connect your electrodes to the alligator clips hanging above the lab jack.
6. Place the beaker with the electroplating solution below the electrodes, and raise it using the lab jack so that the electrodes are about 2/3 covered by the solution. Make sure they don't touch each other or the beaker.
7. Launch the Logger Pro software on the desktop of the lab computer. In the Experiment dropdown tab, select Data Collection and set parameters as follows: mode = time-based, 1 sample/second, and the data collection duration to 1800 seconds.
8. The power supply should already be on. Plug the wire into the terminal while you start data collection on logger pro. This is a source of uncertainty in the experiment, so try and do these 2 things together as closely as possible.
9. The current on the power supply should register as about 0.4 Amps. Adjust it to make it match.
10. **Keep an eye on the amp measurement on Logger Pro. If it drops, swirl your beaker to remove the bubbles that have built up on the screens.**
11. After 30 minutes, stop data collection, unplug the wire from the terminal, but leave the power supply on.
12. Use the integration feature of the Logger Pro software to calculate the total charge Q that was used to plate the electrode. See Equation 2.1.
13. Lower the lab jack, remove the screens, and rinse them with deionized water and acetone before weighing them again to 0.1 mg.
14. Repeat the experiment two more times. You don't need to run the full 30 minutes, but ensure that you deposit at least 0.1 g of copper onto the screen. You can estimate the amount of copper deposited per minute from the results of your first trial. **Be sure to switch which screen is which in between experiments so that you don't completely use up the copper from one of the copper screens.**

Lab Report

Be sure to calculate the following for your lab report:

- Using Avogadro's number, the mass of copper deposited, calculate the total number of electrons transferred. Use the stoichiometry from Equation 2.2 to convert moles of Cu^{2+} to moles of electrons.
- Calculate the charge of 1 electron using the total number of electrons and the total charge (Q).

- Use the known mass-to-charge ratio from J.J. Thompson's experiment and your measured charge of an electron to calculate the mass of a single electron.
- Calculate the average charge and mass of an electron from your three runs and their 95% confidence limits.

Also, include the answers to the following questions in your lab report.

1. Why was it not possible to determine the charge of the electron by this method at the time when Thomson was doing his experiment?
2. A constant current of 0.800 A was used to deposit copper at the cathode of a cell. Calculate the grams of Cu deposited in 15.2 minutes.
3. If the time and current are each uncertain by 1.0% and the mass of copper is uncertain by 0.10%, what is the uncertainty in the calculated electron charge?

Data Sheet

	Run 1	Run 2	Run 3
Mass of Cu Deposited (g)			
Average Current (amp)			
Time (second)			
Total charge delivered (C)			
Mass of Electron (kg)			

Average Charge of Electron (coulombs)	
Standard Deviation	
95% Confidence Limits	
Average Mass of Electron (kg)	
Standard Deviation	
95% Confidence Limits	

3: Behavior of Gasses (Experiment)

Hazard Overview

Chemical Hazards

- Acetone - flammable
- Dry ice - cryogenic
- Liquid nitrogen - cryogenic

Mechanical Hazards

- N/A

PPE

- 100% cotton lab coat
- Splash goggles
- Nitrile gloves
- Work gloves

Dispensary Provided Items

- Dewar
- Glass bulb
- Cardboard lid
- LabQuest with pressure sensor and temperature probe
- 60 mL syringe
- Plastic tubing
- Helium gas
- Liquid nitrogen
- Dry ice
- Ice
- Wooden mallet
- Disposable spatula

Introduction

The study of gases was central to the development of the atomic theory of matter. Understanding the behavior of gases is a fundamental part of modern chemistry. The ideal gas law (Equation 3.1), relates four properties of a gas: pressure P, volume V, temperature T, and the number of moles n.

$$PV = nRT \tag{3.1}$$

This law with the same value of the constant R (universal gas constant) holds approximately for all gases near atmospheric pressure and becomes increasingly accurate as the pressure decreases. In this experiment, you will investigate several corollaries of the ideal gas law. First, you will study how the pressure depends on the volume for a fixed amount of gas at a constant temperature. Equation 3.1 indicates that the PV should remain constant at constant n and T, or that a graph of P versus $1/V$ should be a straight line (Boyle's law). Second, you will examine how P depends on T for a fixed amount at constant V. In this case, Equation 3.1 predicts that a graph of P versus T should be a straight line (sometimes called Gay-Lussac's law). Third, you will determine the temperature in degrees Celsius (°C) that corresponds to absolute zero (0K).

Measurement of Pressure

Mercury (Hg) manometers are particularly important in the measurement of the pressure of gas since they can function as absolute pressure gauges, wherein the pressure is obtained directly from the measured physical characteristics of the instrument. In the case of a Hg manometer, the characteristics are the height (or height difference) of the Hg level(s) and the temperature of the Hg. The temperature is needed to correct for the thermal expansion of Hg from 0°C, which is required to convert the observed reading in mm of Hg to Torr (1 Torr = 1 mm of Hg at 0°C and standard gravity). Near room temperature, this correction amounts to roughly 0.4%. When higher accuracy is required, smaller corrections may also be needed for the expansion of the measurement scale with temperature and local gravity. These will not be significant and thus are not required in your measurements. A variety of other types of pressure gauges are used, but these typically require calibration against an absolute pressure gauge. In this experiment, you will use an electronic pressure gauge that has been calibrated against a Hg manometer.

The electronic pressure sensor or transducer that you will use in this experiment is based on piezo resistance, which describes the change in electrical resistance caused by an applied strain. This type of pressure sensor is widely used, for example, in the aviation (to determine altitude from barometric pressure) and automotive industries (as part of fuel injection systems), among others. A material that demonstrates piezo resistance is implanted or bonded to a thin elastic substrate, often a wafer of single-crystal silicon. The substrate flexes proportional to the differential gas pressure across its two faces, and this transfers a strain to the attached piezo resistor. One side of the substrate in this sensor is exposed to the local atmospheric pressure, while the other side is exposed to the pressure that you want to measure. The electrical resistance of the piezo resistor changes, and the device is designed to have a linear response over the pressure range of the sensor. An output voltage that is proportional to the differential pressure. The output voltage becomes positive or negative as the pressure increases or decreases, respectively, from the local atmospheric pressure. For example, an output voltage of approximately 150 mV occurs if the pressure differs from the local atmospheric pressure by 0.658 atm.

Operation of Lab Equipment

Boyle’s law

1. Prepare the gas pressure sensor and an air sample for data collection by connecting them to the LabQuest Mini module. Open Logger Pro software.
2. In the Experiment dropdown menu, select "Change Units" and change it to "mm Hg."
3. In the Experiment dropdown menu, select "Data Collection." Change the mode to "events with entry."
4. Enter the "Column Name" as "Volume" and the "Units" as "mL." Select OK.
5. Press the green go arrow to begin data collection.
6. To enter the volume data, click "Keep" and input the **total** volume in mL. Select OK to store this pressure/volume data pair.
7. When you are finished collecting data, click Stop.

Experimental Procedure

Measurement of Atmospheric Pressure

Your TA will measure the atmospheric pressure with the barometer located in the room on the wall nearest the dispensary. A barometer is a closed-end Hg manometer that is dedicated to the measurement of atmospheric pressure. It allows a more precise measurement than a simple manometer with a wooden meter stick since the scale is more precise and includes a Vernier so that the Hg height can be measured to 0.1 mm. Your TA will provide instructions on using the barometer, but the basic steps are outlined below, and a good video tutorial can be found here.

1. First, check the level of the liquid Hg at the bottom of the barometer, and adjust the knob on the bottom so that the pyramid point is just barely touching the Hg.
2. Next, use the knob on the side of the barometer to adjust the measurement tool to the height of the mercury.
3. Read the height of the mercury. The markings on the side represent the height of the mercury to the nearest integer. The markers are in the middle of the vernier. Look for the line that lines up perfectly with the scale on the side of the barometer. The marker that lines up is the first digit of the decimal (the tenths digit). These 2 numbers give you your initial measurement.
4. This measurement must be adjusted for the thermal expansion of the Hg. This can be done using the table that is taped to the barometer. Find the temperature and the pressure value in the tables. Subtract this number from your measured mmHg.

Record the Hg height, temperature, and the correction.

Boyle's Law

Goal: Determine the relationship between pressure and volume for a gas with constant number of moles and constant temperature.

The gas we use will be air, and it will be confined in a syringe connected to a Gas Pressure Sensor (see Figure 3.1). When the volume of the syringe is changed by moving the piston, a change occurs in the pressure exerted by the confined gas. This pressure change will be monitored using a Gas Pressure Sensor and Logger Pro. Pressure and volume data pairs will be collected during this experiment and then analyzed. From the data and graph, you should be able to determine what kind of mathematical relationship exists between the pressure and volume of the confined gas. Historically, this relationship was first established by Robert Boyle in 1662 and has since been known as Boyle's law.

The apparatus that you will use to study the dependence of the pressure of a gas on its volume is shown in Figure 3.2. Some notes before you begin:

- Minimize the contact of your hand with the body of the syringe while moving it. This experiment is sensitive to the change in temperature of the gas, and the heat from your hand is enough to change the gas temperature.
- The pressure sensor is also temperature sensitive, so avoid holding it as well.
- You will need to manually maintain the position of the syringe for most settings.
- Be sure to record the temperature of the air near the syringe in your lab notebook.
- To account for the gas that is in the system that's not accounted for by the syringe, add 0.35 mL to your volume measurements.

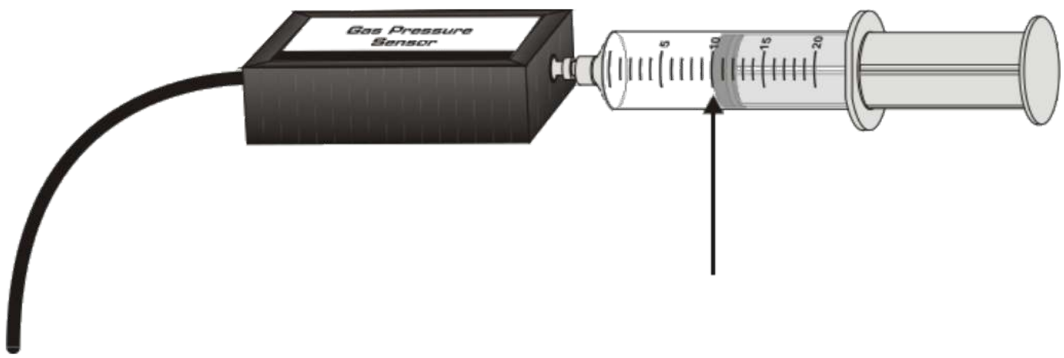


Figure 3.1: Syringe Pressure Setup (CC BY-NC-SA;??????)

The measurement for this part of the experiment is as follows:

1. Set the plunger of the 60 mL syringe at about 30 mL, recording the exact volume to 0.5 mL, and then attach your pressure sensor.
2. **Record the volume** of the syringe from the graduations, aligning the marks with the center of the seal on the plunger that is closest to the end of the syringe with the sensor.
3. **Record the pressure** as reported on the Logger Pro software.
4. **DO NOT INCREASE THE VOLUME ABOVE 50 mL OR BELOW 20 mL.** Doing so can damage the pressure sensor.
5. Hold the syringe in one hand and the plunger in the other while adjusting the volume.
6. Push the plunger in with 2.5 mL increments until you reach 20 mL. Record the pressure at each increment to the nearest 0.1 mL (30.0 mL, 27.5 mL, 35.0 mL, etc.).
7. Next, pull the plunger out from 30 mL in 2.5 mL increments until you reach 50 mL. Record the pressure and volume at each increment. Altogether, you should have pressure measurements between 20-50 mL in 2.5 mL steps **for a total of 13 pressure-volume pairs**.

Gay-Lussac's Law

You will next examine how the pressure of a fixed amount of helium (He) gas in a glass bulb at constant volume varies with temperature. The main reason that He gas is used instead of air is to eliminate problems associated with water vapor. The partial pressure of water vapor in the air can be as high as 20 Torr (0.0263 atm) near room temperature, depending on the relative humidity. This water will condense at the lower temperatures that you will use. Significant deviation from the expected linearity of your Gay-Lussac's Law graph would then occur from higher to lower temperatures since the amount of gas would not be constant. An added benefit of He is that it behaves the most ideally among all gases.

The apparatus that you will use is shown in Figure 3.2. **Have the TA flush and fill your glass bulb with helium gas**. Keep the bulb stoppered with your index finger after filling and before attaching the pressure sensor, and minimize the time the bulb is open to the atmosphere while you attach the pressure sensor.

For this portion of the lab, you will perform 4 pressure and temperature measurements, which will then be used to find the relationship between temperature and pressure. **For the room temperature and the ice bath only**, you will use the thermometer connected to your LabQuest to record the temperature. **DO NOT use the thermometer in dry ice or liquid nitrogen, as you will damage it. These temperature values are already known, and are given below.**

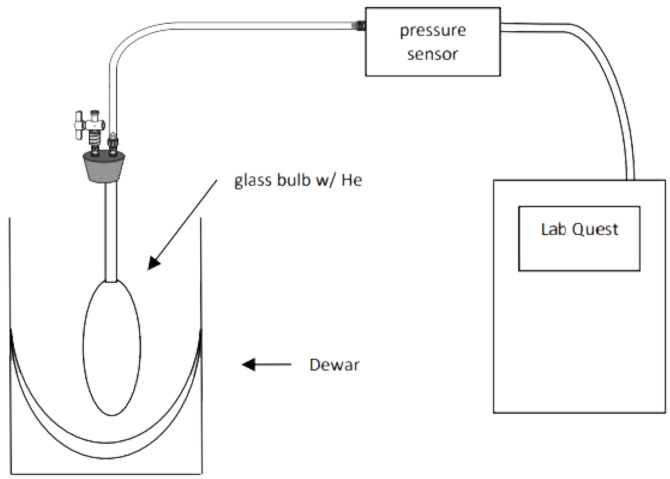


Figure 3.2: Apparatus for Gay-Lussac's Law (CC BY-NC-SA;??????)

Table 3.2: Gay-Lussac Law Measurements

Temperature	Pressure	Bath
		Room Air
		Ice Bath
-78.48°C		Dry Ice + Acetone
-195.8°C		Liquid Nitrogen

Allow the apparatus to equilibrate in a room temperature bath, and then measure the temperature and the pressure reading from the sensor.

The remaining data points will be taken at reduced temperatures, corresponding to the ice point, a "dry ice" (solid CO_2) bath, and liquid nitrogen. These three temperatures are very well-defined and fixed since they result from a phase-equilibrium for a pure substance at constant pressure. The order in which these points are taken is not important. However, the glass bulb should be returned to near room temperature, and its surface dried of condensation before cooling to the next temperature.

Ice Bath

The ice point occurs at 0.00°C and is established when ice and air-saturated water are in equilibrium at atmospheric pressure, namely when both exist in a well-stirred and equilibrated system. Support the glass bulb in a beaker, and pack ice under, around, and over the large volume of the bulb. Add deionized water while replenishing the ice that melts so that the level of the packed ice remains above the water level. The final level of the water should completely cover the large volume of the bulb and roughly the bottom third of the attached capillary to achieve good thermal contact. Stir the bath by moving the bulb up and down slowly. After the output of the sensor has stabilized, record its reading. Note that this may not be at 0°C, which is ok as long as you record the actual temperature.

Dry Ice + Acetone Bath

Disconnect the temperature probe, as it will get damaged based on these parameters. Solid $\mathrm{CO_2}$ ("dry ice") in equilibrium with its vapor at 1 atm provides a temperature of -78.48°C. This equilibrium can be established while also obtaining good thermal contact by using a bath consisting of "dry ice" and a liquid that does not freeze at -78.5°C. The traditional choice of liquid is acetone, an organic solvent $\mathrm{CH_3COCH_3}$.

Caution: Dry ice can cause painful low-temperature "burns" if held in the bare hand for even a few seconds. Use the spatula and gloves provided to handle the pulverized dry ice. Support the dry glass bulb in a dry Dewar flask, and fill the flask about one-half full with acetone. Slowly add tiny quantities of powdered dry ice. The dry ice will at first evaporate almost immediately, and there will be considerable foaming at the surface of the acetone. Add more dry ice only after the foaming has stopped and at a sufficiently slow rate so that the acetone does not foam out of the Dewar and onto the lab bench. Dry ice will begin to accumulate on the bottom of the Dewar only after the liquid has sufficiently cooled. At this time, the dry ice can be added at a greater rate without causing serious foaming. A temperature of -78.5°C is estimated when the bath is well stirred and a slow, steady stream of $\mathrm{CO_2}$ bubbles rises from solid $\mathrm{CO_2}$ at the bottom of the Dewar flask. Record the pressure. **Dispose of dry ice and acetone solution in the proper container in the fume hood.**

Liquid Nitrogen Bath

The equilibrium between a pure liquid and its vapor at the normal boiling point provides a third type of constant temperature bath. You will use liquid nitrogen, $\mathrm{N_2}$, which at 1.00 atm pressure boils at -195.80°C.

Caution: Liquid nitrogen can cause severe low-temperature burns. The liquid also evaporates rapidly when exposed to warm surfaces. Your TA will pour the liquid nitrogen for you. Support the dry glass bulb in a dry Dewar flask so that the top of the flask is roughly halfway up the capillary tube of the bulb. Slowly add small amounts of liquid nitrogen to the Dewar flask. The liquid will evaporate almost instantly. As the flask and the bulb cool, the evaporation rate subsidies and liquid nitrogen will accumulate in the Dewar. The Dewar flask should be filled, and the large volume of the glass bulb should be completely covered. Add the supplied cover to the top of the Dewar flask to minimize condensation from the air into the liquid nitrogen and slow evaporation. After the system has equilibrated, record the output of the pressure sensor.

Lab Report

Be sure to calculate the following for your lab report:

• Report the pressure and temperature of the room on the day of lab.

- Include a **table** with your Boyle's law measurements, a **plot** of P versus $1/V$, and a **linear fit** of the graph to a line. The slope of your line corresponds to nRT, so you should also report the number of moles of gas in your syringe.
- Include a **table** with your Gay-Lussac's law data, a **plot** of P versus T, and a **linear fit** of the graph. Extrapolate the line to $P = 0$ to calculate absolute zero.
- Calculate %error in your value of absolute 0 in temperature. To do this, you will need to convert the value to Celcius, and compare to the literature value of absolute zero (-273.15 K).
- Comment on your sources of error. What were some major causes of error in your experiment, and how can these sources be mitigated in the future?

Also, include the answers to the following questions in your lab report.

1. The density of Hg at 0°C is 13.5951 g/mL. Use the volume expansion coefficient of Hg, $3\alpha = 1.82 \times 10^{-4}\,{}^{\circ}C^{-1}$, to estimate the density of Hg at 20°C.
2. To perform this experiment, what two experimental factors were kept constant?
3. Based on the data and graph that you obtained for this experiment, express in words the relationship between pressure and temperature.
4. Write Boyle's law in terms of P, T, and a constant k. One way to determine if a relationship is inversely or direct is to find the proportionality constant k from the data. If this relationship is direct, $k = P/T$. If it is inverse, $k = P \times T$. Calculate k for the ordered pairs in your data table (divide or multiply the P and T values). How constant were your values?
5. According to this experiment, what should happen to the pressure of a gas if the Kelvin temperature is doubled? Check this assumption by finding the pressure at -73°C (200 K) and at 127°C (400 K) on your graph of the pressure versus temperature. How do these two pressure values compare?

4: Determination of Avogadro's Number (Experiment)

Hazard Overview

Chemical Hazards

- Corrosive sulfuric acid

Mechanical Hazards

- Electrical hazard

Dispensary Provided Items

- 0.5 M potassium sulfate solution
- 100 mL beaker
- Universal indicator
- Nichrome electrodes

Introduction

Several international agreements are important in relating the mass and amount of chemical substances. These involve the atomic mass scale and the definitions of Avogadro's number and the mole. The atomic mass scale is based on the agreement that the relative atomic mass of one atom of ^{12}C shall be exactly 12. A statement that
the atomic mass of ^{1}H is 1.00780 then means that one ^{1}H atom has a mass that is 1.00780/12 times the mass of one ^{12}C atom. This leads to the tabulated atomic masses of elements being dimensionless, abundance-weighted averages of the atomic masses of the various isotopes that make up a naturally occurring macroscopic sample of the element. Avogadro's number, N_0, is defined as the number of atoms in exactly 12 g of isotopically pure ^{12}C. A mole (mol) of any substance is the amount that contains N_0 units of the substance. It follows that the mass in grams of N_0 atoms (one mole) of any element is numerically equal to the relative atomic mass of the element. The same conclusion applies to molecules.

An atomic mass unit (amu or u) is defined as exactly 1/12 of the mass of a single atom of ^{12}C. Since 1 mol of ^{12}C has a mass of exactly 12 g, one atom of ^{12}C has a mass of $12/N_0$ g. Hence, 1 u is numerically equal to the inverse of Avogadro's number. This means that a substance's atomic or molecular mass has the same numerical value, both in u and in g/mol. Another unit of mass that is equivalent to an atomic mass unit is the Dalton (Da). This is used frequently for biological molecules with very large molecular masses (kDa).

The numerical value of Avogadro's number cannot be deduced from purely theoretical considerations but must be determined experimentally by somehow counting the number of entities in one mole. You will do this from data obtained in the electrolysis of a water solution. Electrolysis of H_2O causes it to decompose into the gases H_2 and O_2, with the amounts produced dependent on the number of electrons transferred during the electrolysis. Avogadro's number can be found from the simultaneous determination of the number of moles H_2 and/or O_2 formed and the total number of electrons transferred.

The number of moles of H_2 or O_2 is found from the ideal gas law by collecting the gas and measuring its pressure, volume and temperature. The number of electrons transferred is determined from the measurement of the total charge or number of coulombs that flowed during the electrolysis and the known charge of the electron. One ampere of current flow corresponds to the passage of one coulomb of electrical charge through a circuit in a one-second time interval. Hence, if the current, I, is constant and known, the number of coulombs, Q, can be determined by measuring the time, t, during which the current flows using:

$$Q = I \cdot t \tag{4.1}$$

If the current is not constant, then the integral of $I \cdot t$t must be evaluated to determine Q. To avoid such difficulties, you will perform the electrolysis at a constant current using a power supply.

The electrolysis is carried out by placing two inert electrodes in a H_2O solution and connecting the electrodes to the output terminals of a power supply. When the power supply is turned on and warmed up, a constant current will flow around the circuit. The current is carried in the external part of the circuit (wires) by the movement of electrons and in the H_2O solution by the movement of ions. The negative pole of the power supply delivers electrons to the solution, while the positive pole accepts electrons from the solution. Hence, reduction of the H_2O occurs at the electrode connected to the negative pole, and oxidation of the H_2O occurs at the electrode connected to the positive pole. We can postulate that the electrode processes are

$$2\,H_2O_{(l)} + 2\,e^- \rightarrow H_{2(g)} + 2\,OH^-_{(aq)} \tag{4.2}$$

at the negative pole (cathode, reduction occurs), and

$$H_2O_{(l)} \rightarrow 2\,H^+_{(aq)} + \tfrac{1}{2}O_{2(g)} + 2\,e^- \tag{4.3}$$

at the positive pole (anode, oxidation occurs). The overall reaction is

$$3\,H_2O_{(l)} \rightarrow 2\,H^+_{(aq)} + 2\,OH^-_{(aq)} + H_{2(g)} + \tfrac{1}{2}O_{2(g)} \tag{4.4}$$

The postulated reactions indicate that $OH^-_{(aq)}$ will accumulate around the cathode, making the solution around the cathode basic, while $H^+_{(aq)}$ will accumulate around the anode, making the solution around it acidic. However, if the solution is stirred, the $H^+_{(aq)}$ and $OH^-_{(aq)}$ that form initially at the anode and cathode, respectively, will combine with each other to again make $H_2O_{(l)}$. Hence, the net reaction of the electrolysis in a well-stirred solution becomes

$$H_2O_{(l)} \rightarrow H_{2(g)} + \tfrac{1}{2}O_{2(g)} \tag{4.5}$$

Note that one mole of H_2O is decomposed for every two moles of electrons transferred around the circuit as a current. This follows from the preceding discussion, but it also follows from the net reaction. The oxidation number of each H decreases from +1 in H_2O to 0 in H_2 for a net gain of 2 electrons per H_2O. Concomitantly, the oxidation number of O increases from -2 in H_2O to 0 in O_2 for a net loss of 2 electrons per H_2O. Thus, the net reaction also shows that two moles of electrons are involved for every mole of H_2O decomposed.

Several requirements exist for water electrolysis to proceed at a reasonable rate. Pure liquid water has a very small conductivity since the concentrations of ions, $H^+_{(aq)}$ and $OH^-_{(aq)}$ from its self-dissociation, are very small ($\times 10^{-7} M$). The small conductivity limits the current flow, so an impossibly long time is required for any observable electrolysis to occur. To have significant conductivity and thereby obtain a reasonable current and rate of decomposition, an aqueous ionic solution must be used for the electrolysis. The added ions are chosen to not interfere with the electrode processes. They are present simply to carry current through the solution.

There are also requirements for the voltage that must be applied across the electrodes. This must be sufficient to overcome any energy requirement that may exist for the decomposition to occur. Once this minimum voltage is exceeded, electrolysis occurs. Its rate and the observed current subsequently increase linearly with the applied voltage if there are no complications. Hence, larger values of the applied voltage give larger rates of electrolysis and decomposition. The various parameters are adjusted to get the desired amount of decomposition in a given time. However, care must be taken to avoid excessive heating of the solution from the current flow since this can introduce errors in the data analysis. The details are discussed later in the section that describes the calculations.

You will verify several aspects of the above discussion, along with determining Avogadro's number, in the present experiment. First, you will make some qualitative observations that establish the postulated electrode reactions. Second, you will make quantitative measurements to determine Avogadro's number. Third, you will determine how the rate of decomposition varies with the current.

Experimental Procedure

Qualitative Observations

Figure 4.1 shows schematically the apparatus that will be used for qualitative observation of the electrolysis of H_2O. **Your TA will operate this apparatus (there will be one for the whole class), and you will record your observations.** 0.5 M K_2SO_4 solution will be used to observe the electrolysis of H_2O at a reasonable rate. This solution contains a sufficient concentration of ions, $K^+_{(aq)}$ and $SO^{2-}_{4(aq)}$, to allow this. Several drops of "universal indicator" are added to the K_2SO_4 solution in order to give it a fairly intense color. This indicator changes color depending on the acid/base content of the solution. The changes occur in the same order as the colors in a rainbow (red, orange, yellow, green, blue, indigo, violet) as the solution changes its acid/base concentration by powers of ten from $\times 10^{-4} M$ acid (red) through $\times 10^{-7} M$ acid and/or base (a "neutral" solution) to $\times 10^{-10} M$ base (violet). If you are familiar with the terminology, the indicator color changes like the rainbow as the solution pH goes from 4 to 10.

- Verify that the power supply is set to about 10 V and the current set to a maximum of 0.4 A. Let it warm up for about 5 minutes.
- Add about 50 mL of K_2SO_4 to a beaker, and mix with several drops of universal indicator.
- Fill the funnels with the colored solution. Tap the connecting tube to remove all bubbles from the line.
- Immerse the nichrome electrodes into each funnel and connect the other end to an electrical lead with an alligator clip.
- Connect the leads to the power supply, turn it on (if it's not already on) and observe the changes in the solutions over time.

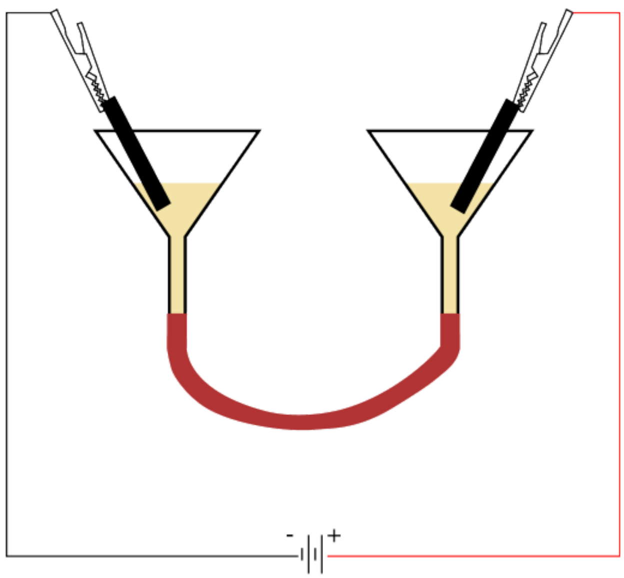


Figure 4.1: Schematic of Experimental Apparatus for Qualitative Electrolysis of H_2O. (CC BY-NC-SA;??????)

Record the color of the K_2SO_4 solution with the added indicator prior to turning on the power supply. Record and interpret what you see happening in each of the two funnels as electrolysis occurs. Make sure to correlate your observations with the polarity of the electrode in the funnel (the polarity of the power supply terminal attached to the electrode). You should be able to ascertain the identity of the gas that is evolved at each electrode from your observations and the postulated electrode reactions given earlier. Periodically check on the solution throughout the lab period.

Quantitative Measurements

For the quantitative measurements, you will collect both the H_2 and O_2 gases that evolve during the electrolysis in inverted burettes. Figure 4.2 shows the apparatus at the start and during the electrolysis schematically. The volumes of the uncalibrated portion of the two burettes between the 50 mL mark and the stopcock, called the "Head Space" in Figure 4.2, need to be known (you measured them in Experiment 1, and should use the value you calculated then).

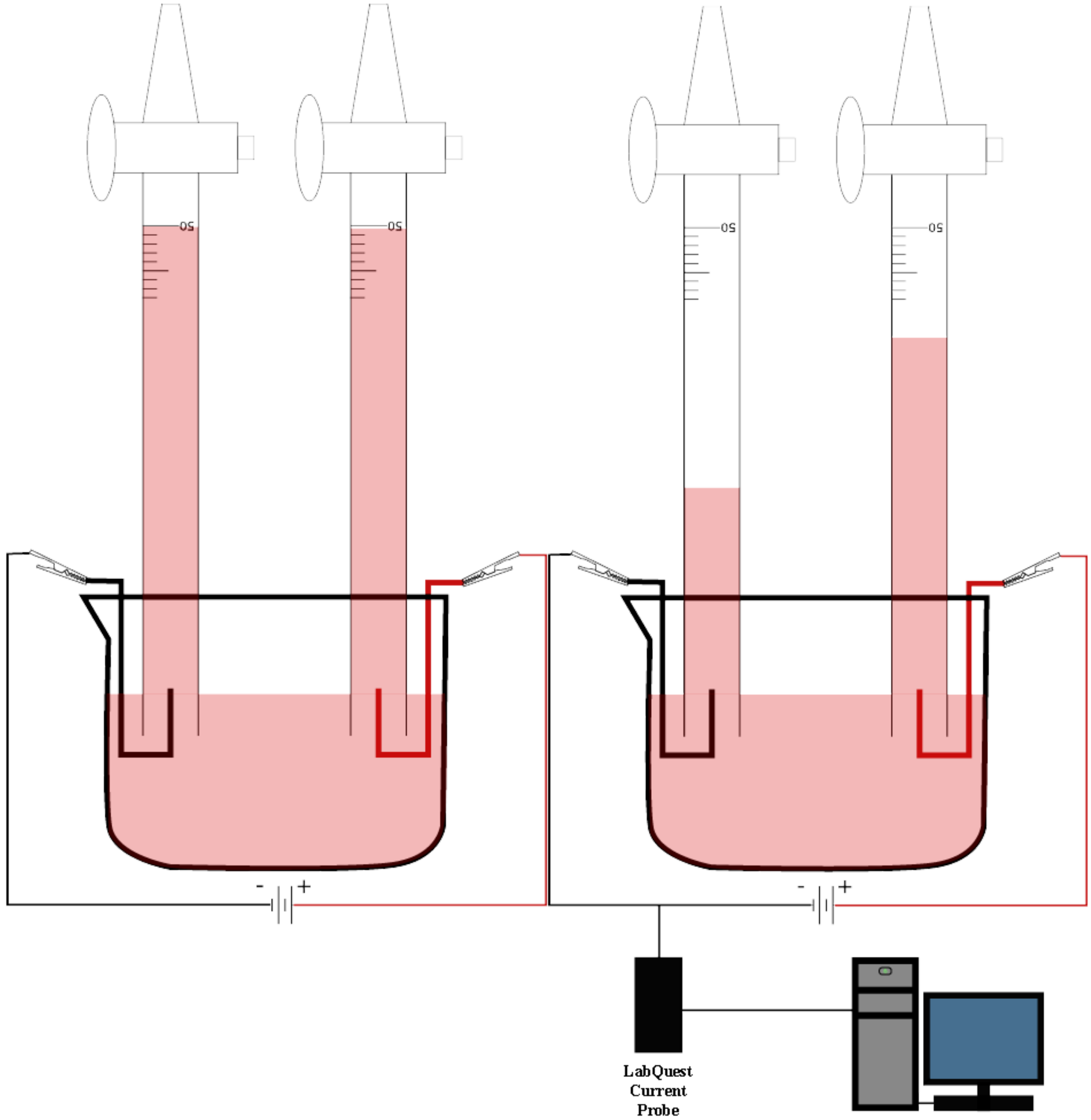


Figure 4.2: Schematic of Experimental Apparatus for Quantitative Electrolysis of H_2O. (The image on the left is before the experiment; the image on the right is what you can expect after electrolysis.) (CC BY-NC-SA;??????)

Measure the pressure with a mercury barometer and record the result. You should check this again at the end of the period and use the average over the period if the value changes. **Consult with your TA for further instructions.**

- Wear gloves and obtain about 400 mL of 0.50 M H_2SO_4 solution in a 600 mL beaker.
- The burette apparatus should already be set up for you at your lab stations.
- Check to make sure that all connections are good. There are two inverted burettes clamped to a support rod. Add a magnetic stir bar into the 600 mL beaker containing the H_2SO_4 solution.
- The power supply should be on and set to the specification: 10 V and 0.4 A.
- Remove the hotplate and place the beaker under the burettes. Place the hotplate back where it was.
- Open the burette stopcock, and carefully draw the dilute H_2SO_4 solution up into the burette using a pipette bulb over the burette tip to near but not above the 50 mL mark. Close the stopcock to keep the solution level stable. Fill both of the burettes.
- Record the starting volume of the solution in each burette, estimating to 0.02 mL and paying attention to the fact that you are reading the burette "upside down."

- Measure with a meter stick to the nearest 1 mm the height of the solution column (the meniscus) from the top of the liquid in the beaker to the top of the liquid in each burette.
- Measure and record the temperatures of the H_2SO_4 in the beaker and the air near the top of the burettes.
- Wait 1-2 minutes and again read the starting volume of the solution in each burette. Make sure that your burette is not leaking.
- Turn on the magnetic stirrer and set the speed of the stirrer to its minimum. Do not spin it rapidly since air bubbles can form and enter the burettes, and gas bubbles from the electrolysis can be pulled out of the burettes. This would introduce errors in your results.
- Under the Experiment dropdown menu on the Logger Pro software, select "Data collection." Set duration to 30 minutes and seconds per sample to 1 sample/min.
- Plug the red wire into the positive terminal and simultaneously start Logger Pro data collection by pressing the "Collect" button. Bubbles should start appearing at the electrodes.
- Run until about 30 mL of gas has been collected in the burette with the greater volume of evolved gas. Unplug the red wire from the positive terminal and record the ending time. **Leave the power supply on.**
- Measure the temperatures of the solution in the beaker and of the air near the mid-height of the burettes. If the electrolysis has caused a significant rise in the temperature of the solution, wait a few (3 to 5) minutes for the temperature of the solution to cool to within 0.5°C of the air temperature.
- Record the final burette readings, and measure the height of the liquid level in the burettes.

Repeat the electrolysis at 6 V, and if time permits, perform a third measurement at 8 V. These measurements will take longer, so you may need to adjust the duration of the electrolysis.

When you are done, drain each burette into the beaker and remove the beaker of solution the same way you put it on. Put the used acid solution back into the original container in the hood.

Thoroughly rinse off the parts of the apparatus and the outside portions of the burettes that were submerged in the acid solution with a deionized water wash bottle into the rinse beaker. Rinse off the magnetic stir bar and return it.

Calculations

You will calculate Avogadro's number on the basis of both the amounts of H_2 and O_2 that were evolved. To do this, you need to compute the number of moles of evolved H_2 and O_2, and the number of electrons that were transferred while the gases were being evolved.

First, determine the average value of the current in ampere (amp, A) and its standard deviation for each run. (Note that even if the observed current did not appear to change over a run, it would nevertheless have a standard deviation corresponding to the minimum current difference that can be read from the ammeter.) Next, multiply the average current by the total electrolysis time in seconds for each run to obtain the corresponding number of coulombs passed through the solution. The resulting product will have the same relative standard deviation as the current, assuming that there is no significant error in the time measurement. Alternatively, you can get the total number of coulombs by integrating the area under the curve recorded by Logger Pro.

Dividing the number of coulombs transferred by the charge of an electron, 1.6021773 $\times 10^{-19}$ C, yields the number of electrons transferred for each run.

An accurate calculation of the moles of H_2 and O_2 that are produced involves a Dalton's Law calculation with several partial pressures. At the start of a run, the gas in the burette volume above the liquid column is a mixture of air and H_2O vapor. The gas volume equals the volume of the uncalibrated region of the burette plus the (positive) difference between the initial burette volume reading and 50.00 mL. The total pressure of this gas mixture is:

$$P_{\text{total in burette}} = P_T = P_{atm} - P_{\text{column height}} = P_{\text{air initial}} \tag{4.6}$$

At the end of a run, the now-larger gas volume in the burette above the liquid column contains air, the evolved H_2 or O_2 gas, and H_2O vapor. The gas volume is equal to the volume of the uncalibrated region and the volume difference between the final burette reading and 50.00 mL. The total pressure of the gas mixture at the end is given by atmospheric pressure minus the pressure corresponding to the new, smaller height of the solution column.

Table 4.1: Vapor Pressure of Pure Water

Temperature / °C	Pressure / Torr
19	16.5

Temperature / °C	Pressure / Torr
20.0	17.5
21.0	18.6
22.0	19.8
23.0	21.1
24.0	22.4

The pressure exerted by the height of the solution column can be calculated from the product of the measured height, the acceleration due to gravity, and the density of the solution.

However, it is simpler to obtain this pressure by comparison with a Hg column at 0°C. An aqueous column with height h and density ρ is related to a Hg column at 0°C with height h_0 and density ρ_0 by:

$$P_{column\ height} = h_0 = h\left(\frac{\rho}{\rho_0}\right) \tag{4.7}$$

The density of Hg at 0°C, ρ_0, is 13.5951 g/mL. The density of 0.50 M H_2SO_4 at 20°C is about 1.030 g/mL; this changes only slightly near room temperature.

The partial pressure of H_2O in the gas mixture is the vapor pressure of H_2O above the solution at the temperature of the solution. Table 4.1 gives the vapor pressure at several temperatures around room temperature for pure water. The correct temperature is somewhat uncertain in the present experiment because of the heating of the solution in the beaker, caused by the current flow, and the minimal mixing of the solution in the burette with the solution in the beaker. You will use the air temperature adjacent to the burette to estimate the temperature of the solution in the burette (and of the gas in the burette). As evident from the data in Table 4.1, if this estimate is accurate to ± 1.0 °C, then the water vapor pressure can be accurate to about ± 1 Torr.

The second concern is the composition of the aqueous solution. Over a 0.50 M H_2SO_4 solution, the vapor pressure of H_2O will be slightly less than that of pure water. The H_2SO_4 dissociates in aqueous solution into $H^+_{(aq)}$, $HSO^-_{4(aq)}$, and $SO^{2-}_{4(aq)}$ ions. These species occupy some fraction of the surface, and thereby reduce the concentration of H_2O molecules at the surface that can escape into the vapor. The vapor pressure of the H_2O is given approximately by the mole fraction of H_2O in the solution times the vapor pressure of pure H_2O (Raoult's Law).

The mole fraction in Raoult's Law is based on the total moles of solute species. H_2SO_4 is dissociated mainly into $H^+_{(aq)}$ and $HSO^-_{4(aq)}$ ions in water solution. Although partial dissociation of $HSO^-_{4(aq)}$ into $H^+_{(aq)}$ and $SO^{2-}_{4(aq)}$ ions also occurs, the amount is small and can be neglected in the present case. The total moles of solute species can be taken as two times the moles of dissolved H_2SO_4.

Therefore, the mole fraction of H_2O is:

$$X_{H_2O} = \frac{n_{H_2O}}{n_{H_2O} + 2n_{H_2SO_4}} \tag{4.8}$$

$$P_{H_2O} = X_{H_2O} \cdot P^*_{vapH_2O} \tag{4.9}$$

The mole fraction of H_2O in a 0.50 M H_2SO_4 solution with a density of 1.030 g/mL is 0.982 on this basis. The vapor pressure of H_2O above the 0.50 M H_2SO_4 solution is then 0.982 times the vapor pressure of pure water. The partial pressure of air changes between the start and end of a run. Since the number of moles of air in the burette is constant, the pressure of air at the end can be calculated from the pressure of air at the start, both the ending and the starting gas volumes, and both the ending and the starting gas temperatures:

$$P_{air} = P_2 = P_1 \cdot \frac{V_1}{V_2} \cdot \frac{T_2}{T_1} \tag{4.10}$$

If the gas temperature is constant, then a simple Boyle's law calculation suffices ($P_1V_1 = P_2V_2$).

The partial pressure of H_2 and O_2 in the respective burette at the end is given by:

$$P_{x_2} = P_T - (P_{H_2O} + P_{air}) \tag{4.11}$$

The moles of H_2 and O_2 that are produced by the electrolysis are calculated for each run from the partial pressure of the evolved gas, the gas volume at the end and the air temperature near the mid-height of the burette at the end.

The partial pressure of H_2 and O_2 in the respective burette at the end is given by the total pressure minus the sum of the vapor pressure of H_2O and the partial pressure of air. The moles of H_2 and O_2 that are produced by the electrolysis are calculated for each run from the partial pressure of the evolved gas, the gas volume at the end and the air temperature near the mid-height of the burette at the end.

Lab Report

Be sure to calculate the following for your lab report:

- Calculate the moles of H_2 and O_2 that are produced in each run.
- Calculate the ratio of moles of H_2 to moles of O_2 for each run, the mean value of the ratio, the standard deviation, and the 95% confidence limits of the mean.
- Obtain Avogadro's number from the data for both H_2 and O_2 for each run, the mean value, the standard deviation, and the 95% confidence limits of the mean. Remember that two moles of electrons are required per mole of H_2O that decomposes.

Examine the results for the three different currents by calculating the moles of H_2 and O_2 that are produced per second at each current. Compare these production rates with the corresponding values for the average current. Discuss your observations.

Also, include the answers to the following questions in your lab report.

1. Calculate the volume in mL of O_2 gas at STP that is generated by quantitative decomposition of 1.000 mL of liquid H_2O at 19.0°C and 25.0°C.
2. Calculate the mole fraction of H_2O in 0.6350 M H_2SO_4, which has a density of 1.0385 g/mL, based on the total moles of solute species and the complete dissociation of the acid into $H^+_{(aq)}$ and $HSO^-_{4(aq)}$ ions.
3. The electrolysis of an aqueous solution generates 50.0 mL of O_2 gas at STP. Assume that the error in the volume is ± 0.3 mL, with no error in the temperature and pressure. What is the number of moles of evolved O_2 and its uncertainty?
4. Assume that the relative error in measuring the volume of the evolved H_2 or O_2 gas is 1.0%, the relative error in the number of electrons transferred is 0.4%, and the relative errors in all other relevant quantities are negligible. What is the relative error in the calculated value of Avogadro's number?
5. Discuss the direction of the error in the calculated value of Avogadro's number that would result in this experiment for the following cases.
6. Estimate the magnitudes of the errors for the following three cases:
   - The vapor pressure of water is entirely neglected.
   - An incorrect density of 1.50 g/mL is used for the H_2SO_4 solution.
   - The thermometer used to measure the temperature of the gas reads low by 1.0°C.

Data Sheet

Outline of Intermediate Calculations

For Each Run

Electrolysis Time (second)	
Average Current (amp)	
Electrons Transferred	
Initial Temperature (°C)	
Final Temperature (°C)	

	H_2 Burette	O_2 Burette
Initial Total Gas Pressure (torr)		
Initial Gas Volume (mL)		
Initial Air Pressure (torr)		
Final Total Gas Pressure (torr)		
Final Gas Volume (mL)		
Final Air Pressure (torr)		
Final H_2 or O_2 Pressure (torr)		
Moles of H_2 or O_2 Produced (mol)		
Avogadro's Number		

Determination of the Avogadro's Number

Barometric Pressure (torr)	

Volume of Head Space (mL)	H_2 Burette:		O_2 Burette:	

	Run 1		Run 2		Run 3	
	H_2 Burette	O_2 Burette	H_2 Burette	O_2 Burette	H_2 Burette	O_2 Burette
Average Temperature (°C)						
Average Current (A)						
Electrolysis Time (s)						
Initial Height of Solution (mm)						
Initial Burette Reading (mL)						
Final Height of Solution (mm)						
Final Burette Reading (mL)						
Moles of H_2 or O_2						

Ratio of Moles of H_2 to O_2			

Average Value	

Standard Deviation	
95% Confidence Limits	

Avogadro's Number in 10^{23}						

Average Value	
Standard Deviation	
95% Confidence Limits	

Rate of H_2 or O_2 (mol/s)						
Rate / Current (mol/A·s)						

5: Measurement of Planck's Constant (Experiment)

Hazard Overview

Chemical Hazards

N/A

Mechanical Hazards

Electrical hazard (power supply)

Dispensary Provided Items

- LED apparatus
- Oscilloscope

Introduction

Planck's constant is one of the fundamental quantities of physics, and the chemical and physical laws of nature are dependent on its magnitude. In this experiment, you will measure Planck's constant by investigating the current-voltage characteristics of and the colors emitted by a series of light-emitting diodes.

Light-emitting diodes (LEDs) are widely used as components in various consumer products. For example, cell phone displays, computer monitors, TVs, and modern lighting all use visible wavelength LEDs. The lighthouse base station used by some VR headsets uses LEDs that emit at wavelengths beyond the visible spectrum (in the infrared). Most commercial LEDs are based on GaAs and GaP, so-called III-V semiconductors, since they are made up of elements from groups III and V of the periodic table. Crystals with the general composition $\mathrm{GaAs_{1\ -x}Px}$, where x varies between 0 and 1, can be produced. The crystals are doped purposefully with small amounts of different impurities in adjacent regions to form a p-n junction or diode. The resulting doped crystals efficiently emit light when a voltage is applied across the junction. The color of the emitted light depends on the exact value of x. For example, the light of wavelength around 950 nm (infrared) is emitted for x = 0, and around 560 nm (green) for x = 1. LEDs that emit at shorter wavelengths are based on SiC. The technology to produce these well-defined, doped crystals under industrial conditions became available only in the late 1960s, and their use in a variety of consumer products has spread rapidly since then.

The operation of an LED depends on the quantized energy levels of the crystal. Let's consider a hypothetical pure crystal containing a large number (roughly Avogadro's number) of identical entities (atoms or molecules). The energy levels (states) of each entity comprising the crystal are identical if the entities are not interacting. Hence, the number of states with a particular energy, or the degeneracy, for the collection of non-interacting entities is very large. Interactions among the entities in the crystal cause these states to have different energies, leading to a spread of energies or an energy band. The details of the band will depend on the electronic structure of the atoms or molecules comprising the crystal and the interactions among these in the crystal. In a pure semiconductor, two bands result from the interaction of the valence orbitals, each containing half the total number of states. The lower band is called the valence band, and the upper one is called the conduction band. The energy difference between the bottom of the conduction band and the top of the valence band is called the band gap, E_g. The electrons in a pure semiconductor cannot assume energies within this forbidden gap. The process is diagrammed in Figure 5.1.

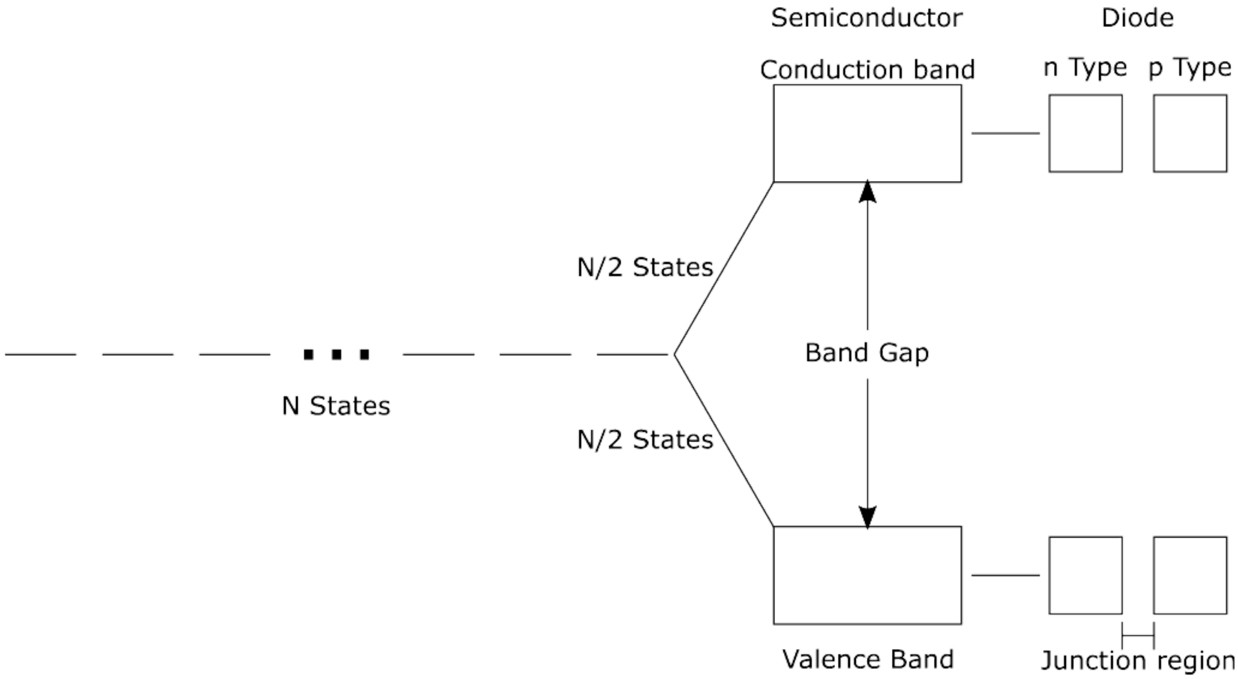


Figure 5.1: Schematic of the Formation of Energy Bands and p-n Junction (CC BY-NC-SA;??????)

The valence band of a pure semiconductor is typically filled, meaning that it contains just the number of electrons that are required to completely fill with opposite spins all of the levels in the band. Its conduction band is essentially empty. In the p region of a diode, the impurity dopant has removed electrons from the valence band, causing vacancies or holes to be produced in the valence band, whereas in the n region, the impurity has added electrons to the conduction band. A barrier exists, however, for the movement of an electron (or a hole) between the n and p regions of the diode. Depending on the particular diode, this barrier is equal to, or slightly lower in energy than, the band gap. If an external voltage that counteracts the barrier is applied across the junction, electrons will flow (be injected) across the junction boundary from the n to the p region.

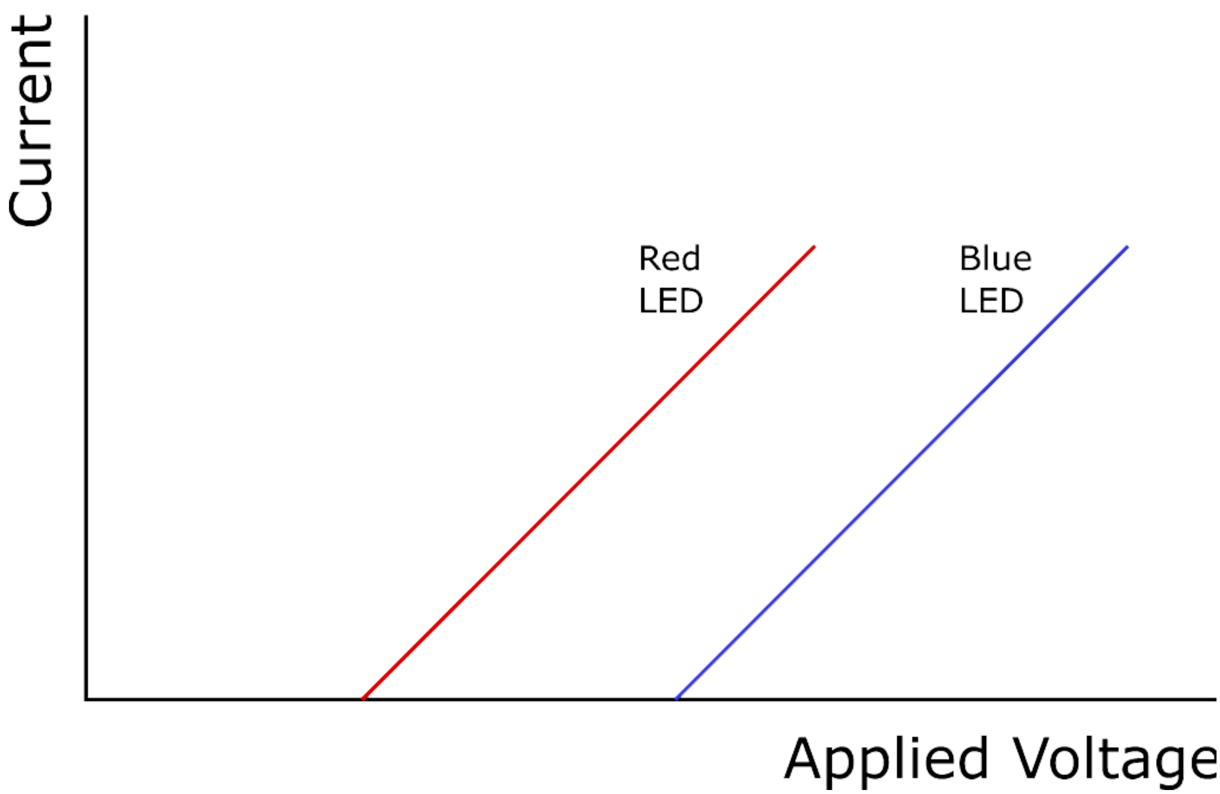


Figure 5.2: Current-Voltage characteristics of LEDs (CC BY-NC-SA;??????)

Figure 5.2 shows the expected current-voltage behavior of an ideal diode. For low applied voltages, there is not enough energy to overcome the barrier between the n and p regions. Hence, no current flows through the diode. When the voltage corresponding to the band gap is reached, the current begins to flow. At even higher voltages, the current increases in proportion to the number of electrons injected into the p region. The behavior shown in Figure 5.2 is reminiscent of the photoelectric effect, except that here the current instead of the kinetic energy of the electrons is measured, and the energy is supplied by an applied voltage rather than light. The energy acquired by the electron from the applied voltage is given by eV (read as electron volt), where e is the electron charge (1.6021773 $\times 10^{-19}$ C) and V is the applied voltage. Since 1 V = 1 J/C, an electron subjected to one volt acquires an energy of 1 eV = 1.6021773 $\times 10^{-19}$ J = 96.48531 kJ/mol. Hence, the voltage at which current begins to flow for a particular diode numerically equals the band gap of the diode in eV.

What happens to the electrons injected into the p region by the applied voltage? These electrons will lose energy by falling into the holes in the lower energy valence band. The released energy appears as heat in a conventional diode. However, in an LED the energy released by some of the electrons falling into the valence band appears as light energy. The color or wavelength of the emitted light depends on the band gap of the material through the Planck-Einstein relation, given by:

$$E_g = h\nu = \frac{hc}{\lambda} \tag{5.1}$$

where h is Planck's constant, ν is the frequency of the light, λ is the wavelength of light, and $c = 299792458\ m/s$ the speed of light in a vacuum. In SI units, the energy is in J, the frequency is in s^{-1} or Hz, the wavelength is in m, and h is in $J \cdot s$. Note that as the applied voltage is increased from zero, the emission of light appears simultaneously with the current flow, since both the light emission and the current flow depend on the injection of electrons into the junction region of the diode. The two phenomena are linked. Plotting the band gap, E_g, against the inverse wavelength, $1/\lambda$, of the emitted light for a series of LEDs should yield a straight line with an intercept of zero and a slope equal to hc.

In the present experiment, you will measure the band gaps for a series of LEDs from their current-voltage characteristics. You will also determine the peak wavelengths of their emissions using a simple spectroscope and fit your data to estimate Planck's constant.

Operation of Lab Equipment

Operation of a Spectroscope

A simple spectroscope uses a grating to disperse light into its various wavelengths. A transmission grating consists of a series of parallel grooves on a flat transparent surface. When light is incident on the grating, each groove acts as a separate source of light waves for the transmitted light, creating interference in the light waves. For constructive interference, the transmitted waves must be in phase with their maxima and minima overlapping in space. Constructive interference occurs when

$$n\lambda = d\sin\theta \tag{5.2}$$

where n is an integer, λ is the wavelength of the light wave, and d is the spacing between the grooves of the grating. The grating acts to disperse the light into its various wavelengths since each wavelength has its maximum intensity at a particular angle, θ, for constant d and a given n. When n = 1, the image of the diffracted light is called the first-order image.

A schematic of a spectroscope is shown in Figure 5.3 Light enters the spectroscope through a slit and exits through a transmission grating to the eye of the viewer. Since the transmitted light undergoes interference, a virtual image of the slit occurs at an angle θ, dependent on the grating spacing and the wavelength of the light. A scale is attached to the spectroscope that allows the wavelength of the light to be read from the position of the first-order image. The width of the image for monochromatic light depends on the width of the slit. The larger the dimensions of the spectroscope and the smaller the grating spacing, the more accurately the position of the image can be determined for a fixed slit width. Research-grade spectrometers use reflection gratings and photoelectric detection of real images. These instruments can exceed several meters.

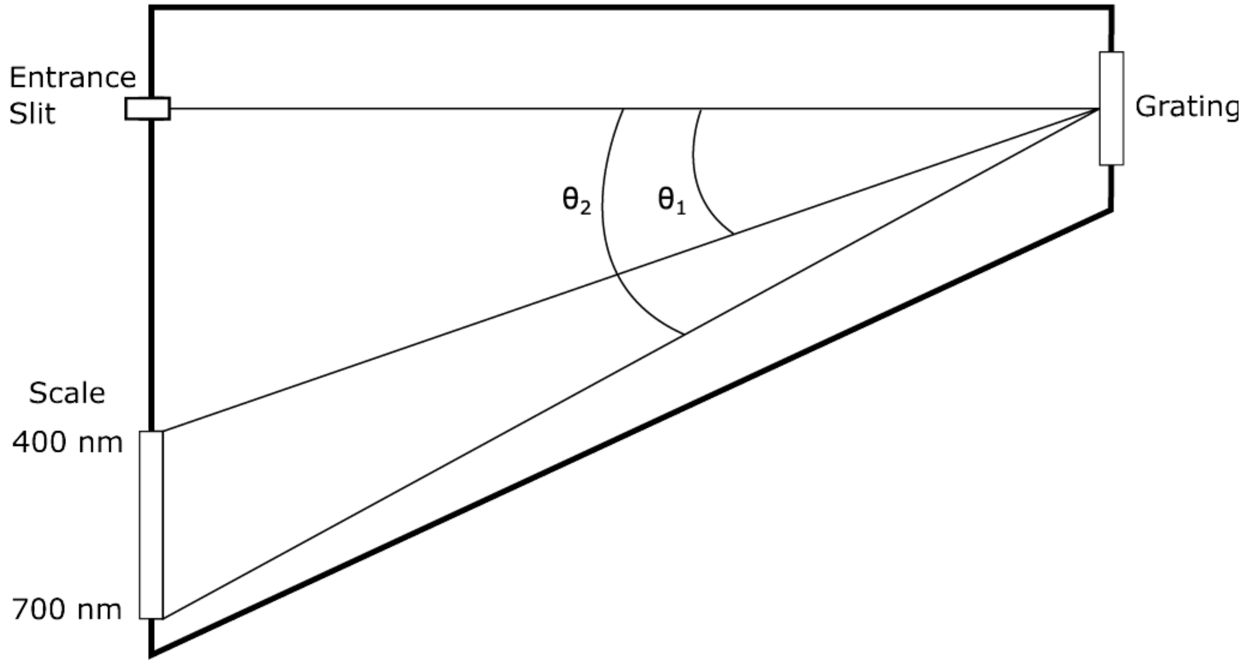


Figure 5.3: Spectrometer Schematic (CC BY-NC-SA;??????)

Experimental Procedure

You will study six different LEDs in this experiment. The wavelengths of their emitted light will be measured using a simple spectroscope for the five LEDs that emit in the visible region of the spectrum, and you will use an oscilloscope to measure their band gap characteristics. The procedure is outlined below.

Figure 5.4 shows a sketch of the apparatus that houses the LEDs with its attached power supply. An individual LED is chosen by the selector switch. Position 1 on the switch corresponds to LED 1 (blue) and position 6 to LED 6 (IR). Intermediate positions on the switch select the other LEDs in order of their emission wavelengths. Position 0 disconnects the LEDs from the circuit. The other control adjusts the maximum applied voltage and, thereby, the intensity of the light emitted by the chosen LED. The two BNC connectors on the apparatus provide two output voltages. The output labeled X gives the voltage that is applied across the LED. The output labeled Y gives a voltage that is proportional to the current through the LED (the current in amps is given by 0.01 times the Y output in volts).

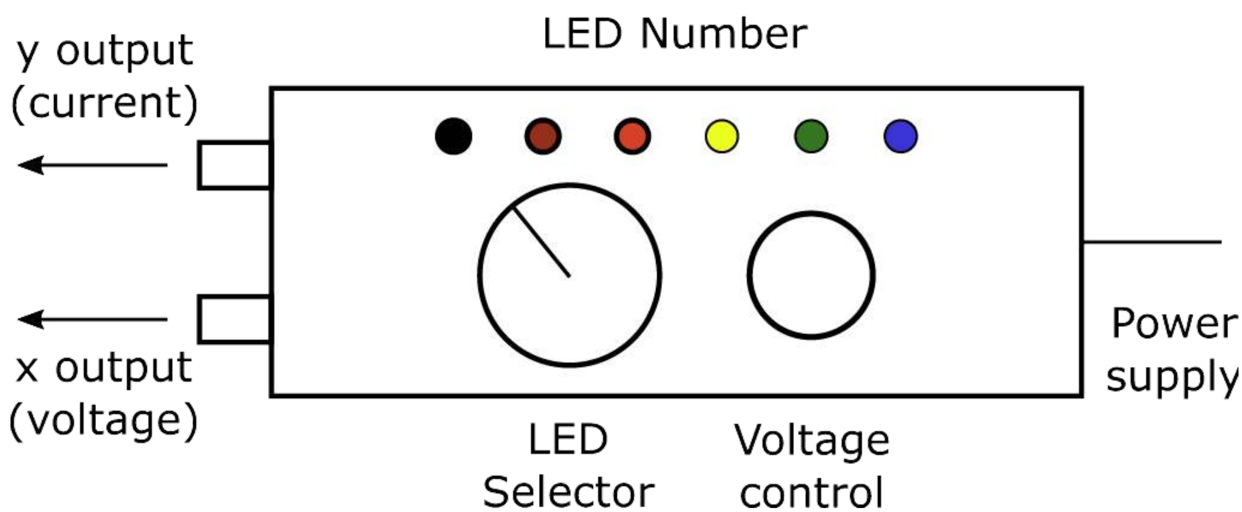


Figure 5.4: Sketch of LED Apparatus (CC BY-NC-SA;??????)

Ensure the power supply is plugged into the line voltage. Do not turn off the power supply throughout the experiment. Switch the selector to position 1 and adjust the voltage control from minimum (fully counterclockwise) to maximum (fully clockwise) while observing the blue LED. Similarly, examine the LEDs corresponding to positions 2 to 5, recording the visual color of each of the LEDs. Then, using a hand-held spectroscope, view in turn each of the lighted LEDs to determine the wavelength of their emitted light. Record the peak wavelength (the wavelength of maximum intensity) for each of the five LEDs 1 to 5, estimating to 5 nm. The peak wavelength for LED 6 is in the infrared at 950 nm, which is beyond the visible range.

You will next use an oscilloscope to deduce the band gaps of the LEDs. A schematic of an oscilloscope is shown in Figure 5.5 with its inputs and major controls identified. The oscilloscope has two inputs that are labeled CH 1 or X and CH 2 or Y (CH stands for channel). When the Sec/Div switch is set to X-Y, voltages at the X and Y inputs will cause the oscilloscope to show a Y (vertical) versus X (horizontal) trace on its display. The volts per division along the Y and X axes are set by the Volts/Div switch for the corresponding axis. The position on the display corresponding to zero volts for the Y input is adjusted by the Position control above the V/Div switch for the Y input. The position for the X input is adjusted by the control above the Sec/Div switch, not by the corresponding control for the X input. The intensity of the trace and its focus are adjusted by the controls adjacent to the display above the power-on switch. All controls you need for this experiment are highlighted on the scopes you will use.

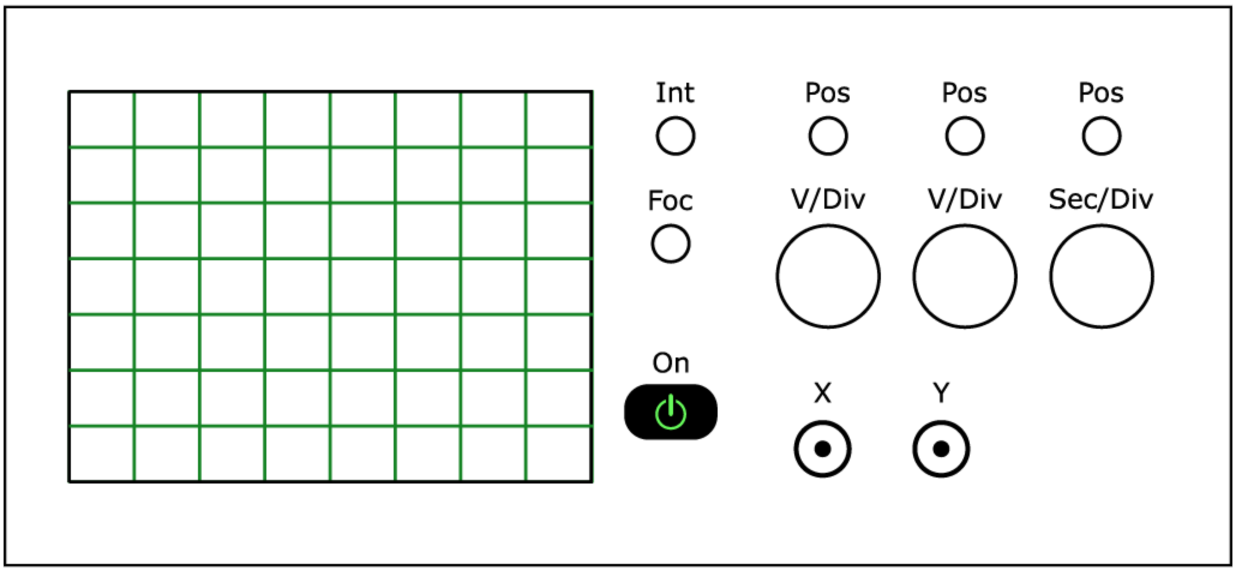


Figure 5.5: Schematic of an Oscilloscope with Major Controls Labeled (CC BY-NC-SA;??????)

Set the oscilloscope Sec/Div switch to X-Y. Ground the X and Y inputs by sliding the switches labeled AC-GND-DC (located directly above the inputs) to GND. Then, move the position of the spot on the display to the intersection of the center horizontal graticule and the first vertical graticule from the left of the display. This sets the origin or the position of zero volts on the display. Slide the switches to DC to make measurements.

Set the selector switch of the LED apparatus to position 0, adjust its voltage control to minimum, and connect its X and Y outputs to the X and Y inputs, respectively, of the oscilloscope with electrical cables. Then, select LED 1 and slowly turn the voltage control to its maximum on the LED apparatus, while observing the display on the oscilloscope. You should obtain a trace on the display similar to one of the traces in Figure 5.2. The trace should be a horizontal line from zero volts to a voltage slightly less than the band gap, a transitional curving portion just before the band gap, and a positively sloping line at higher voltages. Note that the LED lights at the voltage corresponding to the onset of the positively sloping portion of the trace, namely when current flows through the diode. Visually extrapolate the linear, sloping portion of the trace to intersect the extension of the horizontal portion in order to estimate the voltage corresponding to the band gap of the LED. The setting of the V/Div switch for CH 1 or X gives the volts per line along the horizontal axis (the minor ticks along the middle graticules are at 0.2 divisions). Similarly, estimate the band gaps of the other LEDs, including the LED at position 6, whose emission is in the infrared. The band gaps are in the range of 1-3 eV, so the horizontal sensitivity should be in the range of 0.2-0.5 volts per division, in order that the origin (zero volts) and the onset of the sloping portion are simultaneously visible in the display. You should use the most sensitive setting possible for each LED, to get the best estimate of its band gap.

You may have to adjust the volts/div when switching to a different LED to be able to accurately measure the band gap.

The vertical sensitivity should be adjusted for ease in the extrapolation of the sloping line to the horizontal axis (approximately 0.2-1.0 volts per division). Before you record your estimate for the band gap, it is prudent to again ground the X and Y inputs to check that the position of zero volts has not moved.

Lab Report

Be sure to calculate the following for your lab report:

- Graph the band gaps in eV for the six LEDs against the **inverse of their peak wavelengths** in nm. Fit the data to the equation for a straight line, and calculate Planck's constant in J s from the slope. Note that the y-intercept is 0, if it is not, you should force the intercept through zero.
- Report your graph, your best estimate for Planck's constant in J s, % error in Planck's constant, and its standard deviation in J s.

Also, include the answers to the following questions in your lab report.

1. Compare your result for Planck's constant with the accepted value of 6.626076 $\times 10^{-34}$ J s. Discuss possible reasons for any disagreement.
2. A LED emits light with a wavelength of 940$\pm$5 nm. What is the band gap and its error in eV?
3. Light with a wavelength of 800 nm is incident on a transmission grating that has 1200 grooves per mm. Calculate the diffraction angle θ (see Figure 5.3) for the first order image.
4. A potential problem when using diffraction gratings is that the images for different wavelengths can occur at the same diffraction angle in different orders. What other wavelengths, and in what order, have images that occur at the same diffraction angle as that for 800 nm light in the first order?

Data Table

LED Number	Visual Color	Wavelength (nm)	Band Gap (eV)
1			
2			
3			
4			
5			
6	IR	950	

Planck's Constant (J s)	
Standard Deviation (J s)	

5: Measurement of Planck's Constant (Graph)

Key points:

- Units are especially important to pay attention to in this lab.
- We will be comparing your experimental Planck's constant with the actual literature value 6.63E-34.

Summarized Procedure:

1. Obtain the wavelength and band gap of six different LEDs using a simple spectroscope and oscilloscope, respectively.
2. For each wavelength, take the inverse of it and change the units from cm^{-1} to m^{-1}. For each recorded band gap, convert the units from eV to Joules.
3. Make 2 graphs. The first is energy (eV) vs. the inverse of wavelength in cm^{-1}. The second is energy (Joules) vs. the inverse of wavelength in m^{-1}.
4. The slope and y-intercept values of each graph's line of best fit and the values' respective standard deviations will be automatically calculated.
5. Using the slope from the energy (J) vs. inverse of wavelength (m^{-1}), the experimental value of Planck's constant (J*s) can then be found.

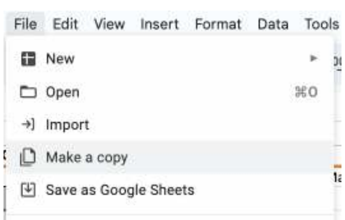
Calculations:

The main formula that is used in this is Planck's Constant (h) = $\frac{\lambda E_g}{299792458}$. λE_g is found obtained from the graphs. 299792458 is the speed of light in meters per second.

Making the Graphs:

Example Graph	https://docs.google.com/spreadsheets...f=true&sd=true
Student Graph (MAKE A COPY OF THIS, FILL IT OUT WITH YOUR DATA TO MAKE YOUR GRAPHS)	https://docs.google.com/spreadsheets...f=true&sd=true

Procedure for Graphs:



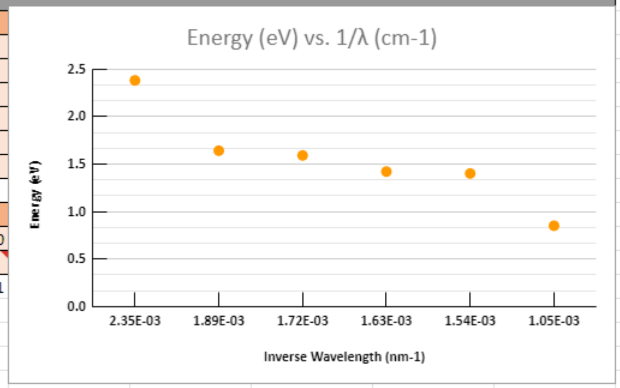
Data			
Input:			
LED Number	Visual Color	λ (nm)	Bandgap (eV)
1	blue	425	2.38
2	green	530	1.64
3	yellow-orange	580	1.59
4	light red	615	1.42
5	dark red	650	1.4
6	infrared	950	0.85

Output:				
LED Number	1/λ (cm-1)	λ (m)	1/λ (m-1)	Bandgap (J)
1	2.35E-03	4.25E-07	2.35E+06	0
2	1.89E-03	5.30E-07	1.89E+06	0
3	1.72E-03	5.80E-07	1.72E+06	0
4	1.63E-03	6.15E-07	1.63E+06	0
5	1.54E-03	6.50E-07	1.54E+06	0
6	1.05E-03	9.50E-07	1.05E+06	0

Graphs

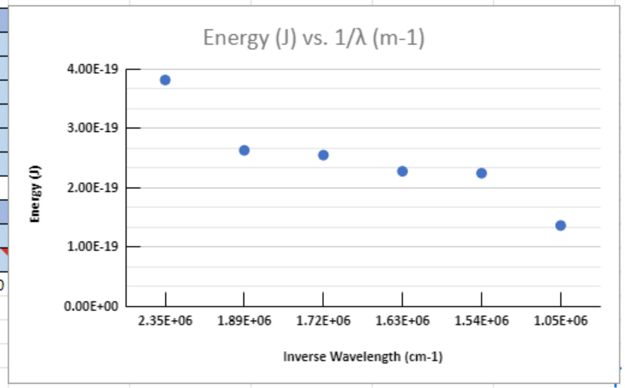
LED Number	1/λ (cm-1)	Bandgap (eV)
1	2.35E-03	2.38
2	1.89E-03	1.64
3	1.72E-03	1.59
4	1.63E-03	1.42
5	1.54E-03	1.4
6	1.05E-03	0.85

Energy (eV)	slope	y-intercept
value	923.3249271	0
stdev	28.79042279	#N/A
	0.9951621675	0.1227829531



LED Number	1/λ (m-1)	Bandgap (J)
1	2.35E+06	3.81E-19
2	1.89E+06	2.63E-19
3	1.72E+06	2.55E-19
4	1.63E+06	2.28E-19
5	1.54E+06	2.24E-19
6	1.05E+06	1.36E-19

Energy (J)	slope	y-intercept
value	1.48E-25	0.00E+00
stdev	4.61E-27	#N/A
	9.95E-01	1.97E-20



Results	
Planck's Constant (J*s)	
Experimental Value	4.93E-34
Standard Deviation	1.54E-35
Actual Value	6.63E-34

1. Click "Student Garph" link above.
2. Click "File" then "Make a Copy.
3. Following the instructions on YOUR BLANK COPY, add your experimental data obtained from the lab into the YELLOW HIGHLIGHTED BOXES. The rest of the sheet is set to run the calculations for you. Do NOT adjust the other parts. If you do, you can go back to the original and obtain the necessary functions to fix it.
4. The output part below your data input has data that is used in the calculations. Do not edit this part of the sheet.
5. Your graphs will automatically show up in this part of the sheet. The colors of the lines and dots on the graph are coordinated with the results on the left side. On the left, you will find the points that have been graphed. Also displayed will be its slope, y-intercept, and standard deviations. The #N/A is correct as there is no standard deviation for the y-intercept. Again, do not edit this part of the sheet.
6. Here are your results for the lab! The experimental value is found using the slope of the Energy (J) vs. inverse of wavelength (m^{-1}).

6: Optical Spectroscopy of Atoms (Experiment)

Hazard Overview

Chemical Hazards

Mechanical Hazards

Dispensary Provided Items

- Spectroscope
- 1 test tube rack
- Forceps
- 5 disposable test tubes
- Sharpie
- 6 nichrome wires
- Unknown (1/student)

Introduction

The most important evidence for quantization or discrete energy levels comes from the analysis of spectra, namely the light emitted or absorbed by atoms and molecules. In this experiment, you will observe and assign part of the emission spectrum of electronically excited hydrogen atoms. You will also observe the emission of light by several other excited gaseous species, and the absorption of light by a colored species in solution.

When the light emitted or absorbed by gaseous atoms is dispersed by a prism or grating, it is found to consist of discrete lines having characteristic colors or wavelengths, λ. These λ's correspond to the energy differences between the quantized energy states of the atom. The relationship between the energy, E, of a light quantum or photon and its frequency or wavelength, λ, is given by:

$$E = h\nu = \frac{hc}{\lambda} \tag{6.1}$$

where $h = 6.6206076 \times 10^{-34}\ Js$ is Planck's constant and $c = 2.99792458 \times 10^{8}\ m/s$ is the speed of light. Thus, the energy differences between the quantized states of an atom are readily determined experimentally by measurement of the wavelengths of its emitted light. We will consider two cases in more detail.

The simplest atom is hydrogen. As might be expected, it has the simplest energy level diagram and the simplest spectrum. The electronic energy levels, E_n, of the hydrogen atom, from either primitive Bohr theory or the quantum theory, are given by

$$E_n = \frac{-C}{n^2} \tag{6.2}$$

where n is the principal quantum number and C is a collection of constants. The quantum number, n, can assume any positive integer value from one to infinity, with the value of infinity corresponding to the separated proton and electron and to an energy of zero. The energy difference between the two states of the hydrogen atom becomes:

$$\delta E = E_i - E_f = C\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \tag{6.3}$$

where $n_i > n_f$. When the hydrogen atom undergoes a transition from a higher-lying initial state with n_i to a lower-lying final state with n_f, the energy given by Equation 6.3 is liberated and appears as light. Using Equations 6.1 and 6.3, the wavelength of the emitted light is given by:

$$\frac{1}{\lambda}=\frac{C}{hc}\left[\frac{1}{n_f^2}-\frac{1}{n_i^2}\right]=R_H\left[\frac{1}{n_f^2}-\frac{1}{n_i^2}\right] \tag{6.4}$$

where R_H is the Rydberg constant. Different spectral series result from Equation 6.4 if n_f remains fixed while n_i varies from n_f+1 to infinity. These series are named after the scientists who discovered them: Lyman for n_f = 1, Balmer for n_f = 2, Paschen for n_f = 3, and Brackett for n_f = 4. Figure 6.1 shows a schematic energy level diagram for the hydrogen atom and the transitions associated with some of the spectral series. Each series occurs in a particular wavelength range.

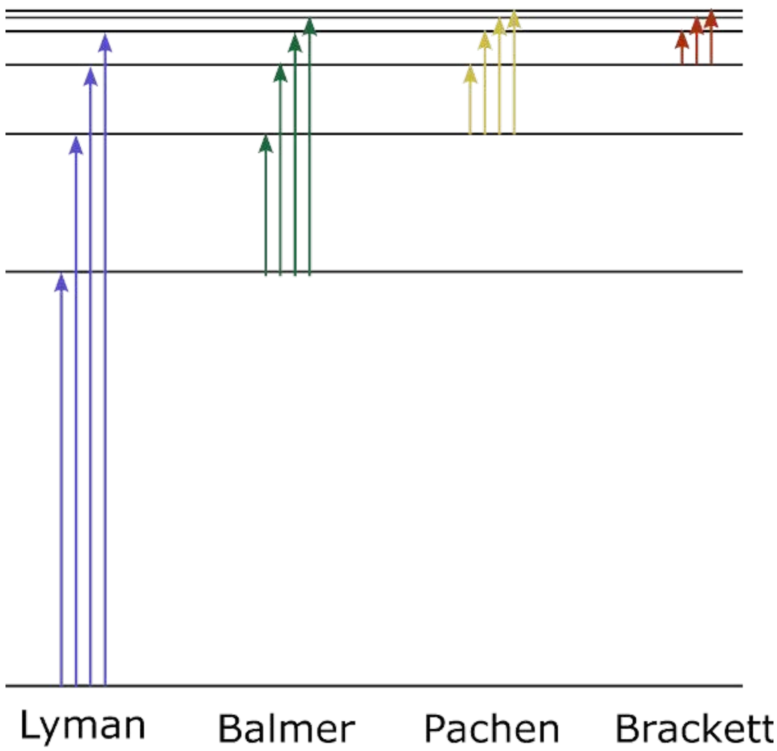


Figure 6.1: Energy Level Diagram and Spectral Series of Hydrogen (CC BY-NC-SA;??????)

The emission spectrum of the hydrogen atom can be generated by running a high-voltage discharge through a tube that contains gaseous hydrogen molecules. In the discharge, hydrogen molecules are knocked apart by bombardment with energetic electrons. The fragments contain electronically excited hydrogen atoms which can emit light as they relax to lower energy states and ultimately to the ground state. The discharge lamp is designed to maximize the emission of light by excited hydrogen atoms. This requires that the rate of recombination of excited atoms is not too fast. Two colliding atoms cannot recombine to form a stable molecule unless a third body is also present to carry away any excess energy. At the low pressure (about 1 Torr = 0.00131 atm = 0.132 kPa) in the discharge tube, such three-body collisions are sufficiently infrequent in the gas phase that most of the recombination occurs at the wall, which serves as the third body. Since water inhibits hydrogen-atom recombination on glass, a small amount of water has been added to the discharge tube to help maintain a high concentration of hydrogen atoms. Discharge lamps that emit atomic spectra are available commercially for a variety of atoms that can be generated from gaseous precursors.

The energy level diagrams and spectra of many-electron atoms are more complicated. The major reason for this is that the energies now depend on both the principal quantum number, n, and the angular momentum quantum number, l. Additional complications result if the electronic configuration involves a partially filled subshell with several electrons since different occupations and spins for the electrons within the subshell lead to states of different energies (Hund's rules). We will consider the lithium atom, which has the simple ground-state electron configuration of $\mathrm{ls}^2 2\mathrm{s}^1$. The lower-lying excited states of lithium result from the promotion of the 2s valence electron to the higher energy orbitals.

Not all possible transitions between the energy levels of an atom can occur through the absorption or emission of light. Rather, selection rules operate and limit the allowed transitions to those wherein $\Delta l=\pm 1$. For example, the 2s – 2p transition is allowed ($\Delta l=\pm 1$), whereas the 2s – 3s transition is forbidden ($\Delta l=\pm 0$). Since the energies in the hydrogen atom depend only on n, with the possible values of l having the same energies for a given n, the $\Delta l=\pm 1$ selection rule is not apparent in the spectrum of the hydrogen atom. The relative intensities of the spectral lines in the emission spectrum of lithium depend on the relative concentrations of lithium atoms whose valence electron has been excited to the various higher energy orbitals.

The atomic spectrum of most metallic elements cannot be obtained using a simple discharge tube, since most metals have too low a vapor pressure to support a discharge. Mercury is a notable exception. Conventional fluorescent lamps involve a discharge through mercury vapor. The light emitted by excited mercury atoms is absorbed by a phosphor that coats the inside wall of the fluorescent lamp. The absorbed light excites the phosphor, which then emits light. The solid phosphor acts to convert the discrete line spectrum emitted by the gaseous mercury atoms into a broad band of "white" light. Part of the mercury line spectrum does escape the fluorescent tube. This is readily seen by viewing a fluorescent light with a simple spectroscope. Non-discharge type lamps, and other means of excitation, can be used to observe the atomic spectra of other metals.

You will generate excited alkali and alkaline earth metal atoms using the heat of a Bunsen burner flame. When certain metal salts are heated in a flame, the salts vaporize and dissociate. Gaseous metal atoms in excited states are formed by the reduction in the flame of the metal cation to the metal. In the case of lithium, the only excited atoms present in sufficient concentration for visual detection of their emission in the flame are those whose valence electron is in a 2p orbital. Hence, only the line at 671 nm, which corresponds to the allowed 2s – 2p transition, is seen. Similarly, only select lines are seen for the other metal atoms that you will investigate using a flame source. This is due in part to the relatively low temperature of the Bunsen burner flame, and in part to the transmission properties of the grating in the spectroscope and the sensitivity of the eye. Other lines can be recorded using spectrophotometers together with alternate means of excitation.

Experimental Procedure

Mercury Spectrum

You will first measure the wavelengths of several of the emission lines from excited mercury atoms in the fluorescent lights in the lab. Aim a spectroscope at a fluorescent light. You should see three or four lines on top of the continuous "white" background emission from the phosphor. The colors of these lines are violet, blue, green, and yellow. You may not be able to observe the violet line above the continuous background, since it is weak and at the edge of the visible range. Record the wavelengths of each of the observed lines. After you are comfortable with the use of a spectroscope, and determining the wavelengths of any observed lines, the remaining three parts of this experiment can be done in any order.

Flame Spectra

The emission from other excited metal atoms will be observed using a Bunsen burner as a flame source. Aim the spectroscope so that the hottest part of the flame is just below the bottom of the slit. One of the team members dips a nichrome wire into a solution containing a dissolved salt of the metal and then places the wet wire in the hot part of the flame. A second member observes and records the wavelength(s) of the emitted line(s). The emission of light by the excited metal atoms lasts only for a few seconds until the solution is consumed. Repeated observations will be necessary to measure the exact wavelength(s) for each solution. Look carefully for weak lines in the violet region, since these are easily overlooked.

Label four test tubes (Li^+, Na^+, Ca^{2+}, Sr^{2+}) and place about one mL of each standard solution in the appropriate test tube. Contamination is a problem, so do not return any solution to the standard bottles. Obtain a length of nichrome wire for each standard and for your unknown. Twist a small spiral loop at one end of each wire and heat the looped end in a Bunsen burner flame until the yellow sodium flame fades. Dip the looped end of the wire in dilute HCl and repeat the heating until no sodium flame is observed. Assign a treated wire to each standard solution and your unknown solutions. Do not exchange these wires during the experiment, or contamination will result. Also, do not place the wire on the bench top; always put the wire back into the solution assigned. Another possible source of contamination in these measurements is the sodium present in the detergent. Do not wash any of the containers with detergent, since the last traces will be virtually impossible to rinse away.

For each solution and wire, the looped end of the wire is dipped into the solution and then placed into the hottest part of the flame. The flame is viewed simultaneously with the spectroscope to measure the wavelength(s) of the emission line(s). Record for each standard solution the color and wavelength of the line(s). Each student then measures
and records the wavelength(s) emitted by his/her unknown and, by comparison with the wavelengths determined for the standard solutions, identifies the metal ion(s) present in his/her unknown.

They are two compounds in your unknown. Do not dispose of any solutions until the unknowns are determined.

Absorption Spectrum of Aqueous Co_2^+

A source of continuous intensity over a range of wavelengths is needed to observe absorption spectra. The light from this source is transmitted through a sample, and the reduction in intensity of the transmitted light at particular wavelengths recorded. A convenient source of continuous white light is an ordinary tungsten light bulb, which contains an electrically heated tungsten filament that radiates as a near blackbody.

View a tungsten lamp with a spectroscope. You should see the continuous spectrum of visible colors that are emitted by the lamp. Insert a square cuvette containing a stock aqueous solution of Co_2^+ between the tungsten lamp and the spectroscope. A range of wavelengths will be absorbed by the solution. Estimate and record this range by viewing the transmitted light with a spectroscope. Sharp lines are not seen in fluid solution, or for gases at high densities, due in part to the range of perturbations in the energy levels caused by the range of intermolecular interactions (different intermolecular distances and orientations).

Balmer Spectrum of Hydrogen

The emission spectrum of atomic hydrogen in the visible region is measured using a discharge lamp. View the lamp with a spectroscope and record the wavelengths of the emitted lines. You should be able to see three lines (blue-violet, blue-green, and red) with ease. The fourth, the violet line, may require patient adaptation of the eye, since it is close to the edge of the visible range and easily overlooked.

Lab Report

The observed atomic hydrogen lines belong to the same series. From a graphical analysis of the wavelength data, you can obtain the value of the Rydberg constant, R_H, the quantum number of the final state, n_f, involved in all four of the observed transitions, and the four values of the quantum number of the initial state, n_i. Plotting $\frac{1}{\lambda}$ against $\frac{1}{n_i^2}$ allows the data to be fit to Equation 6.4 with a slope (m) of $-R_H$ and an intercept (b) of $\frac{R_H}{n_f^2}$. However, this result will be obtained only for the correct sequence of integer values for n_i.

Be sure to calculate the following for your lab report:

- Report the transition assignments of your hydrogen series, the value of Rydberg constant. To find the correct sequence of n_i's, use a spreadsheet to make plots of $1/\lambda$ in cm^{-1} for the four observed lines against $\frac{1}{n_i^2}$ for the following three trial sets of values: n_i = 2, 3, 4, 5; n_i = 3, 4, 5, 6; and n_i = 4, 5, 6, 7. The wavelengths are associated with the n_i's in inverse order for each set of values (red line with the smallest n_i and violet line with the largest n_i). The sequence of integers that gives the best straight line gives the correct assignment. Determine the slope of this line by least-squares analysis. Obtain the Rydberg constant in cm^{-1} from the slope, n_f (as a floating point number, not an integer) from the values of the slope and intercept, and the standard deviations in both parameters. Round n_f to an integer and assign integer values of the quantum numbers n_i and n_f to each of the observed lines in the hydrogen atom spectrum.
- Turn in printouts of your graph, showing all three plots of your data and the best-fit line for the most linear plot, and of your spreadsheet. Include in your spreadsheet the calculated value of n_f as a decimal number (not rounded to an integer) and its standard deviation.

Also, answer the following questions in the report.

1. Calculate the wavelengths of the three lowest energy lines of the Paschen and Brackett series of the hydrogen atom. Do any of these occur in the visible region of the spectrum?
2. Ionization of the H atom from its ground state corresponds to the transition $n_i = 1 \longrightarrow n_f = \inf$. Light with this or greater energy will ionize the H atom, with any excess energy going into the kinetic energy of the ejected electron. Calculate the kinetic energy in J and the speed in m/s, along with the respective uncertainties, of the ejected electron if the H atom is ionized by light with a frequency of $5.00 \pm 0.03 \times 10^{15}\ s^{-1}$.

Data Tables

Mercury Spectrum

Color	Wavelength (nm)

Flame Spectra

Ion	Color(s)	Wavelength (nm)
Li^{+}		
Na^{+}		
Ca^{2+}		
Sr^{2+}		

Unknown Number	Observed Wavelengths	Ions Present

Absorption Spectra

Wavelength Range (nm)

Balmer Spectra

Color	Wavelength (nm)	n_i	n_f

Fitting Results

	Value	Percent error
Rydberg Constant		
Calculated n_f		

LibreTexts

6: Optical Spectroscopy of Atoms (Graph)

Key Points

- Three graphs will be produced for various n_i values.
- The quantum number of the final state, n_f can be found using the slope and intercept of the most accurate graph.

Summarized Procedure

1. Obtain the wavelengths of each emitted line in nm.
2. Take the inverse of each wavelength and convert to the units of cm.
3. Three graphs will be made plotting $\frac{1}{\lambda}$ against $\frac{1}{n_i^2}$ for the following three trial sets of values: n_i = 2, 3, 4, 5; n_i = 3, 4, 5, 6; and n_i = 4, 5, 6, 7. The wavelengths are associated with the n_i's in inverse order for each set of values (red line with the smallest n_i and violet line with the largest n_i).
4. The sequence of integers that gives the best straight line gives the correct assignment. The slope of the line is determined by least-squares analysis. In the provided spreadsheets, the slope and y-intercepts are calculated for you as well as the standard deviations in both parameters.
5. The quantum number of the final state, n_f. can be found in from the slope and intercept. Refer to the calculations for further information.

Calculations

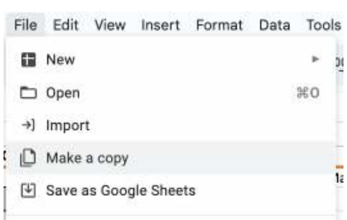
Two main formulas that will be used to obtain n_f are the slope of the line equaling $-R_H$ and the intercept equaling $\frac{R_H}{n_f^2}$.

Plugging in the values obtained from the selected graph, we can then solve for the quantum number of the final state.

Making the Graphs:

Example Graph	https://docs.google.com/spreadsheets...f=true&sd=true
Student Graph (MKAE A COPY OF THIS, FILL IT OUT WITH YOUR DATA TO MAKE YOUR GRAPHS)	https://docs.google.com/spreadsheets...f=true&sd=true

Procedure for Graphs:



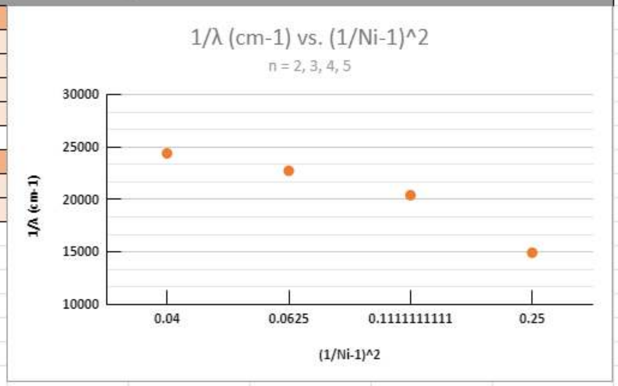
Data		
Input:		
Number	Visual Color	λ (nm)
1	violet	
2	blue-violet	
3	blue-green	
4	red	

Output:	
Number	1/λ (cm-1)
1	24390.2439
2	22727.27273
3	20408.16327
4	14925.37313

Graphs

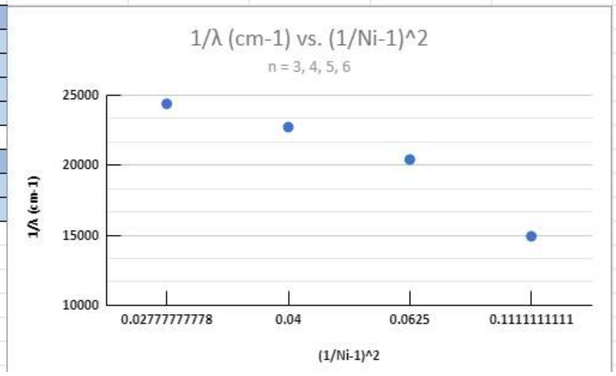
Ni-1	(1/Ni-1)^2	1/λ (cm-1)
5	0.04	24390.2439
4	0.0625	22727.27273
3	0.1111111111	20408.16327
2	0.25	14925.37313

	slope	y-intercept
value	-43629.1174	25669.49916
stdev	2943.350499	417.1671859



Ni-2	(1/Ni-2)^2	1/λ (cm-1)
6	0.02777777778	24390.2439
5	0.04	22727.27273
4	0.0625	20408.16327
3	0.1111111111	14925.37313

	slope	y-intercept
value	-112220.4463	27384.95547
stdev	2354.714793	160.6716908



Results	
	slope
value	-112220.4463
stdev	2354.714793

1. Click "Student Graph" link above.
2. Click "File" then "Make a Copy."
3. Following the instructions on YOUR BLANK COPY, add your experimental data obtained from the lab into the YELLOW HIGHLIGHTED BOXES. The rest of the sheet is set to run the calculations for you. Do NOT adjust the other parts. If you do, you can go back to the original and obtain the necessary functions to fix it.
4. The output part below your data input has data that is used in the calculations. Do not edit this part of the sheet.
5. Your three graphs will automatically show up in this part of the sheet. The colors of the lines and dots on the graph are coordinated with the results on the left side. On the left, you will find the values plotted with respect to their n_i value. There will also be each graph's respective slope value, y-intercept, and standard deviations. Again, do not edit this part of the sheet.
6. Here are your results for the lab! The slope and its standard deviation will be for the graph that produced the most linear line. You will now use this data accordingly to solve for the quantum number of the final state. The data is simply consolidated to this table for your ease. Again, do not edit this part of the sheet.

7: Quantitative Spectrophotometry and Beer's Law (Experiment)

Hazard Overview

Chemical Hazards

Mechanical Hazards

Dispensary Provided Items

Introduction

The absorption of electromagnetic radiation is the basis of numerous modern analytical methods. These methods can provide, for example, qualitative information concerning molecular formulas and bonding, and quantitative information regarding the energy levels and the concentrations of the absorbing species. In this experiment, you will compare the visible absorption spectra in an aqueous solution of several transition metal cations, examine how their separate absorptions depend on the concentrations of the ions in the solution, and use the combined absorptions to determine the concentrations of these ions in an aqueous mixture.

The visible region represents only a small portion of the electromagnetic spectrum. The types of transitions (quantized energy levels) that are associated with the different regions of the electromagnetic spectrum are summarized in Table 7.1. For example, the absorption of light in the visible region is typically due to the promotion of valence electrons to higher energy orbitals. In the visible spectra of transition metal cations, d-orbitals often are involved, as predicted from their electron configurations.

Table 7.1: Types of Spectroscopic Transitions

Type of Transitions	Portion of Spectrum	Approximate Wavelengths
Operations of Spins	Radio Waves	≥ 10 cm
Molecular Rotations	Microwaves and Radar	100 - 1 mm
Molecular Vibrations	Infrared	1000 - 1 μm
Valence Electrons	Near IR, Visible* and Near UV	1000 - 200 nm
Core Electrons	Far UV and X-Rays	2000 - 1 Å
Nuclear Transitions	Gamma Rays	≤ 1 Å

* Visible Region: $\lambda = 390$ nm to 700 nm

The amount of light that a species absorbs in a spectroscopic transition can be related quantitatively to the number of absorbing species. This relationship is called the Beer-Lambert Law, or more simply Beer's Law. Consider monochromatic light of a given intensity incident on a sample, as shown in Figure 7.1. If this light can be absorbed by the sample, then the transmitted light will have a lower intensity than the incident light. The transmittance, T, is defined as the ratio of the transmitted intensity, I, to the incident intensity, I_0:

$$T = I/I_0 \tag{7.1}$$

The percent transmittance, , is simply 10^2T. The transmittance is decreased if either the concentration, c, of absorbing species is increased or the path length, b, of the sample is increased since both will increase the number of absorbing species in the path of the light.

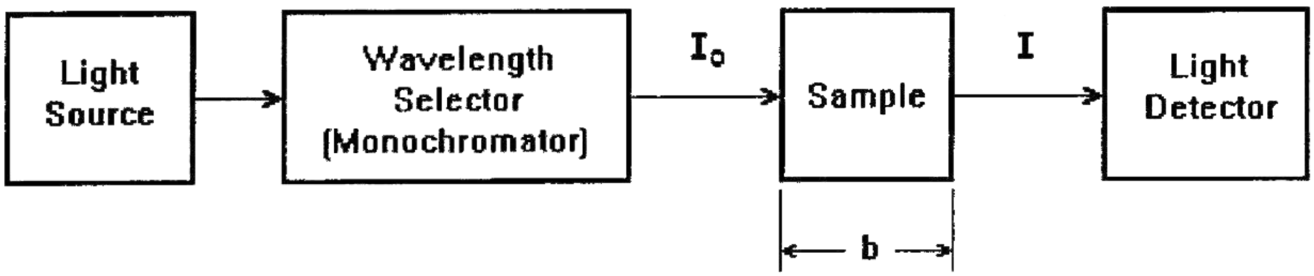


Figure 7.1: Schematic Diagram of the Spectrophotometric Absorption of Light (CC BY-NC-SA;??????)

A related quantity is the absorbance, A, which is given by

$$A = -\log_{10} T \tag{7.2}$$

The absorbance is particularly important since it, and not the transmittance, is directly proportional to the concentration of the absorbing species and the path length. This proportionality constitutes Beer's Law and is commonly written as

$$A = \epsilon b c \tag{7.3}$$

The path length, b, is usually expressed in cm (usually 1 cm), and the concentration, c, in molarity of moles per liter (M). The quantity ϵ is called the molar absorptivity and has units $\mathrm{M}^{-1}\mathrm{cm}^{-1}$ so that the product ϵbc is dimensionless, as are both the absorbance, A, and the transmittance, T. The molar absorptivity is characteristic of the species. It tells us how much light the species absorbs at a particular wavelength. A graphical plot of either the absorbance at constant path length or the molar absorptivity versus wavelength is called the absorption spectrum of the species.

The logarithmic relation between the transmittance and concentration requires justification. This arises since in each infinitesimal portion of the path the decrease in the intensity is proportional to the intensity incident on that portion. As the light travels through the sample, the intensity decreases in each succeeding portion are smaller since the intensity reaching that portion is less. The mathematical details are given at the end of this experiment's instructions.

Beer's Law forms the basis for the analytical use of spectroscopy to determine concentrations. As indicated by equation 7.3, a plot of the absorbance at a given wavelength for a particular species versus the concentration of the species yields a straight line with a slope equal to ϵb and an intercept of zero. Since the path length of the cell used for the absorbance measurements is typically known, the molar absorptivity of the species at a chosen wavelength is readily determined from such a Beer's Law plot. The concentration of the species in an unknown sample can then be determined by measuring the absorbance of the sample at the same wavelength in any cell of known path length.

The wavelength for such an analysis should be chosen so that small changes in wavelength do not yield large changes in absorbance. Namely, the chosen wavelength should be in a relatively flat portion of the absorption spectrum. Typically, a wavelength associated with a maximum in the spectrum is chosen, since at a maximum the slope of the spectrum is zero or horizontal and simultaneously good sensitivity is obtained in the analysis (significant absorbance for a given concentration).

One should always establish, before its analytical use, that Beer's Law is followed by a species over the concentration range of interest since deviations from Beer's law often occur at high concentrations. Typically, these deviations can be traced to changes in the absorbing species or the bulk solution with concentration. For example, in concentrated solutions the solute molecules are closer together on average and interact with each other, changing their energy levels and spectroscopic properties from a dilute solution. The species of interest also may exist in equilibrium with other species that have different molar absorptivities. In such cases, a graph of absorbance versus concentration will appear to deviate from a straight line at high concentrations.

If two species are present, and neither affects the light-absorbing properties of the other, then the observed total absorbance is simply the sum of the absorbances of the individual species. When this is true, the individual concentrations can be determined from spectrophotometric measurements. Because interactions often do arise, sometimes when least expected, the absorption spectra of the species should be investigated when they are separate and when they are simultaneously present to determine whether the absorbances are indeed additive before any analytical spectrophotometric measurements.

Two cases can be differentiated. Consider the case where at some wavelength one of the species absorbs strongly, while the other does not absorb, and at some other wavelength, the converse is true. Then, these two wavelengths can be used separately to determine the concentrations of the two species, just as if the other component were not present since the absorbance at each of the wavelengths depends on the concentration of only one of the two species. The case where both species contribute to the absorption at the two wavelengths is somewhat more complicated, but the two concentrations can still be determined if the absorbances are additive.

Let's use the same path length b for all measurements. Then, the parameters ε and b in equation 7.3 can be combined into a single parameter k that depends only on the species and the wavelength at constant path length. The absorbance of species 1 at a particular wavelength, which we label by the subscript i, can be written as

$$A_{1i} = k_{1i}c_1 \tag{7.4}$$

The total absorbance at this wavelength for a solution of the two components becomes

$$A_i = A_{1i} + A_{2i} = k_{1i}c_1 + k_{2i}c_2 \tag{7.5}$$

A similar relation holds for any wavelength, so long as the absorbances are additive. Hence, the measurement of the total absorbance at two wavelengths (i = 1 and 2) provides equation 7.6 and equation 7.7, in the two unknown concentrations c_1 and c_2:

$$A_{\mathbf{1}} = A_{1\mathbf{1}} + A_{\mathbf{21}} = k_{1\mathbf{1}}c_1 + k_{\mathbf{21}}c_2 \tag{7.6}$$

$$A_{\mathbf{2}} = A_{1\mathbf{2}} + A_{\mathbf{22}} = k_{1\mathbf{2}}c_1 + k_{\mathbf{22}}c_2 \tag{7.7}$$

The various ks are determined from Beer's Law plots for the separate components at the two particular wavelengths chosen for analysis. Simultaneous solution of equation 7.6 and equation 7.7 then yields the concentrations of the two absorbing species, c_1 and c_2, in a mixture that contains both in terms of the absorbances of the mixture at two wavelengths, A_1 and A_2, and the ks.

Operation of Spectrometer

A spectrophotometer to measure absorption spectra combines a light source that emits a continuous band of radiation, a monochromator (usually a reflection grating with associated optics) to select a narrow range of wavelengths, and a photoelectric detector to measure the light intensity, as indicated in Figure 7.1. The optical path of a typical spectrophotometer is shown in Figure 7.2.

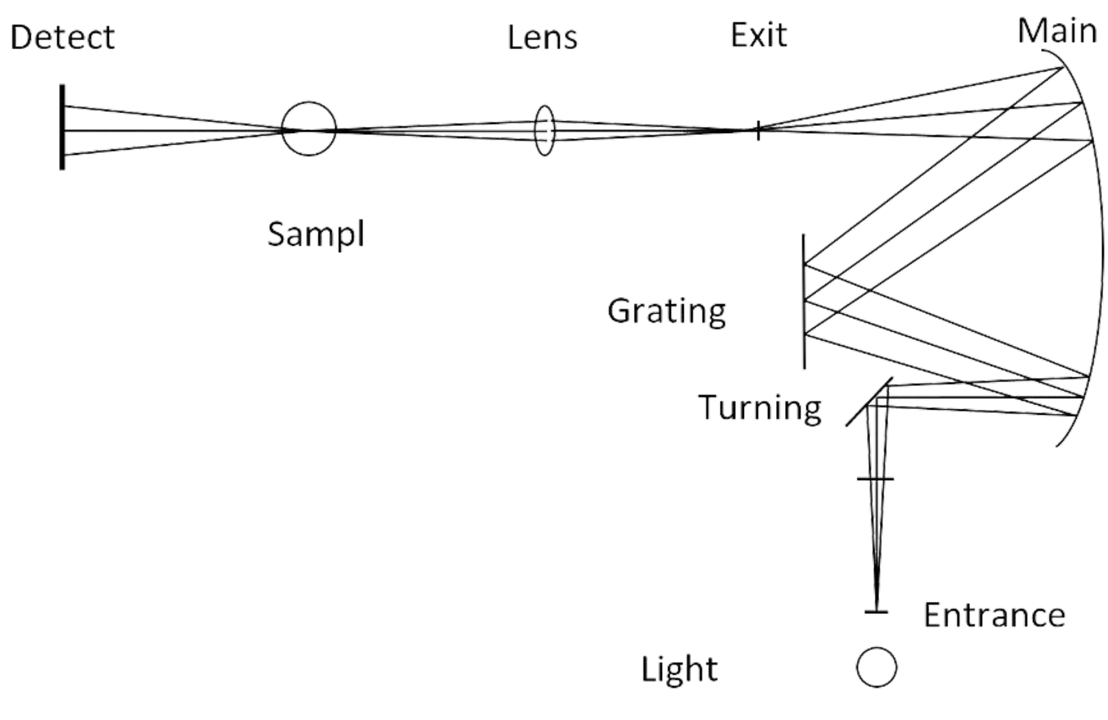


Figure 7.2: Schematic Diagram of the Optical Path of a Spectrometer (CC BY-NC-SA;??????)

In this experiment, you will be using an Ocean Optics USB2000 spectrophotometer that can measure light over a wide range of wavelengths simultaneously. The light path in this instrument is shown in Figure 3. A color picture of the light path is posted on the laboratory wall. (Note that instead of using optics to select a narrow range of wavelengths, all wavelengths are measured at the same time by the detector, a linear CCD array. CCD stands for charge-coupled device, which is the same kind of electronic device used in digital cameras and scanners.)

The light source (not shown in Figure 7.3) is a violet LED-boosted tungsten bulb (390-900 nm) integrated into a package including the cell holder that attaches directly to the spectrometer. The light from the lamp passes through the cell and then enters the spectrometer through an entrance slit. The light is then reflected from a collimating mirror and is dispersed by a diffracting grating. The resulting dispersed light is reflected from a focusing mirror and then strikes the solid-state CCD detector that generates an electrical signal proportional to the radiant power (light intensity) at each wavelength. The signal from the detector is then transmitted to a computer where software plots the spectrum on the display in transmittance or absorbance units.

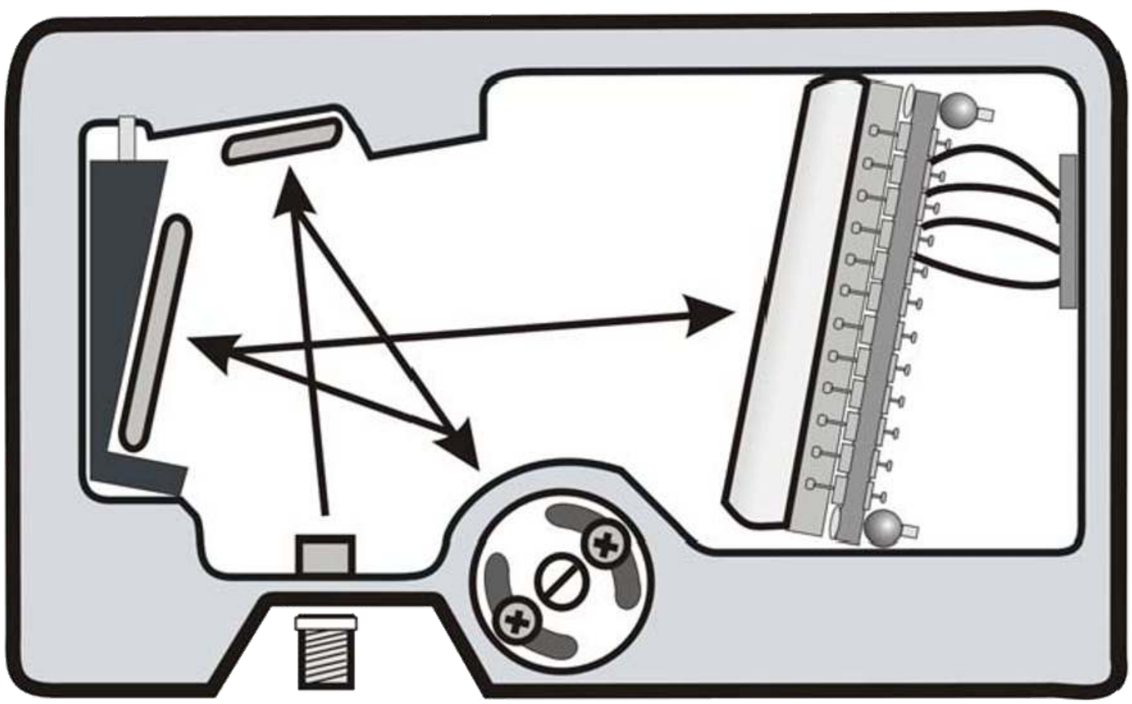


Figure 7.3: Drawing of Ocean Optics USB2000 Spectrometer (CC BY-NC-SA;??????)

The calibration procedure entails setting 100%T over the wavelength range with a cuvette containing a reference or blank sample. This is equivalent to setting absorbance to zero. This is required since the output of the lamp and the sensitivity of the detector vary with wavelength. The software automatically sets absorbance to zero (100%T) and stores the reference spectrum. The blank solution is missing the component of interest but is otherwise as identical as possible to the solution to be analyzed for the component of interest. Often the component of interest is simply dissolved in a solvent while the blank solution is just solvent. An identical cuvette containing the solution of interest is then inserted into the spectrometer, and the absorbance is shown over the wavelength range on the monitor. The reading for the solution then represents the absorbance at the chosen wavelength due to the component of interest. The calibration has accounted for any absorption (or reflection or scattering) of the light by the cuvette and other species in the reference solution.

The proper handling of a cuvette is important. Often one cuvette is used for reference and a second cuvette for the sample. Any variation in the two cuvettes (smudges or scratches on the surface, etc.) will cause errors. There are several important precautions to follow:

1. Do not handle the lower portion of the cuvette through which the light passes.
2. Wipe off any liquid drops or smudges on the lower portion of the cuvette with a clean, supplied Kimwipe. Do not use rough paper (which may scratch the cuvette) or cloth towels (which may leave lint).
3. When using two cuvettes simultaneously, always use one for the reference or blank and the other for various samples. **Do <u>not</u> interchange the cuvette.**
4. Use the cuvette that has the lower absorbance for the reference. This can be established by filling both cuvettes with deionized water and measuring the absorbance of each cuvette.
5. Insert the cuvette into the spectrometer carefully to avoid any possible scratching of the plastic surface.
6. Always rinse the cuvette with several small portions of the solution before filling it with the solution.

Procedure

Refer to the Logger Pro with Spectrophotometer section in Appendix 3 and calibrate your spectrometer with the blank solution (DI water). To measure the spectrum of a standard or unknown solution relative to the reference, rinse and fill the cuvette 3/4 full with the solution of interest and insert it into the sample holder of the spectrometer. Ensure that the frosted side of the cuvet is orthogonal to the direction of the incident light.

Note: In the following parts, **DO NOT dispose of any waste solutions of Cr^{3+} or Co^{2+} by pouring down the drain.** Collect these wastes during the laboratory period in your 600 mL beaker. At the end of the laboratory period, pour the accumulated waste into the waste container provided in the hood. Rinse your beaker with a small amount of water and transfer the rinse liquid to the container. Your rinsed beaker can then be stored in your locker.

Absorption Spectra of Aqueous Cr^{3+} and Co^{2+}

- Obtain around 40 mL each of aqueous stock solutions that are 0.0500 M in Cr^{3+} and 0.15 M in Co^{2+}, respectively.
- You will measure the absorption spectrum of each of these solutions from 400 to 900 nm.
- In each case, rinse the sample cuvette first with deionized water and then with a small amount of the solution you wish to measure, discarding the rinse into your waste beaker. Then, fill the cuvette 3/4 full of the solution. Store the remaining solution for later use.
- **Record the maxima.** These should be around 575 nm for Cr^{3+} and 510 nm for Co^{2+}. These will be the wavelengths that are used for recording absorbance for each dilution.

Dilution Prep for Cr^{3+}

Prepare solutions that are 0.0100, 0.0200, 0.0300, and 0.0400 M in Cr^{3+} by dilution from the 0.0500 M stock solution. The dilutions are carried out using 2 and 4 mL volumetric pipettes and a single 10 mL volumetric flask. You should be able to perform each dilution with only one or two transfers with the pipets. Carefully rinse the pipettes and the flask with deionized water before their use to remove any residue from earlier use. Blow out the remaining water in each pipette, and rinse each with a small amount of the stock solution. Discard the rinse stock solution into your waste beaker. To minimize error, it is best to start by making the least concentrated dilution first.

- Prepare a table that shows the required volume of stock solution to prepare each dilution using $M_1V_1 = M_2V_2$.
- For example, the 0.01 M (M_2) solution is prepared by pipetting 2 mL (V_1) aliquot of the 0.05 M (M_1) stock solution into the 10 mL (V_2) volumetric flask.
- After the proper amount of stock solution has been transferred to the volumetric flask, add deionized water with a squeeze bottle to bring the total volume to the horizontal mark on the volumetric flask.
- Using parafilm cover the flask, and mix the solution well by inverting the flask about 15 times.
- Transfer the solution from the volumetric flask to a labeled test tube for storage and cover the top of the test tube with a piece of Parafilm.
- Rinse the volumetric flask several times with a small amount of deionized water, discarding the waste into your waste container. Prepare the remaining dilutions by changing M_2 and resolving for V_1.

Dilution Prep for Co^{2+}

- Following the same procedure, prepare 0.0300, 0.0600, 0.0900, and 0.1200 M solutions of Co^{2+} from the 0.1500 M stock solution.
- You will need to calculate the required volume of stock solution to pipette into the 10 mL volumetric flask.

Analyzing the Dilutions

- Starting with the least concentrated solution, measure the absorbance of each Cr^{3+} and Co^{2+} dilution at the two wavelengths that you determined earlier.
- Measure the absorbance for each dilution at both wavelengths before changing the solution.
- Then, empty the sample cuvette into your waste container, rinse it with a small amount of the new solution, pour the waste into your waste beaker, fill the cuvette with the new solution, and measure its absorbance at the two wavelengths.
- Record the absorbances at each of the two wavelengths and the concentration.
- After lab, you will use a spreadsheet to prepare 2 Beer's Law plots of absorbance versus molar concentration for Cr^{3+} and Co^{2+} respectively. For each species, perform a least-squares fit of the data for each plot and obtain the product of the molar absorptivity and the path length for each species and wavelength from the slopes. These correspond to the various ks in equations 7.6 and 7.7. Include the best-fit lines on the graphs.

Analysis of an Unknown Cr^{3+} and Co^{2+} Mixture

- Obtain an unknown Cr^{3+}:Co^{2+} sample from your TA and record its number.
- Measure the absorption spectrum of your unknown, and determine its absorbance at the two analysis wavelengths, following the earlier procedures.
- Clean up your station, disposing of the chemical waste in the appropriate waste beaker. All glass waste should go in the glass disposal.

Calculations

- After lab, you will use a spreadsheet to prepare 2 Beer's Law plots of absorbance versus molar concentration for the two ions. For each species, perform a least-squares fit of the data for each plot and obtain the product of the molar absorptivity and the path length for each species and wavelength from the slopes. These correspond to the various ks in equations 7.6 and 7.7. Include the best-fit lines on the graphs.
- From the two absorbances at the analysis wavelengths and the ks, **calculate the concentration in moles per liter of each of the components in the unknown** by setting up a pair of simultaneous equations analogous to equations 7.6 and 7.7 and solving for the two unknowns. Graph the absorption spectrum of your unknown, restricting the wavelength range from 350 to 750 nm. On the same graph, show the expected spectra of the individual components for the concentrations that you determined and the sum of these expected spectra. The expected spectra are obtained from your measured spectra for 0.05 M Cr^{3+} and 0.15 M Co^{2+} through Beer's Law by scaling for the concentration since the path length is constant. For example, if the Cr^{3+} concentration in your unknown is x M, then the expected absorbance from Cr^{3+} at any wavelength is $x/0.05$times the absorbance of the 0.05 M solution at this wavelength for a constant path length. The expected absorbance of the unknown mixture is then the sum of the absorbances of the individual components at each wavelength. Your absorption spectrum for the unknown mixture should closely agree with the calculated sum of the individual spectra since the absorbances are additive in the studied system.
- Give the results of the analysis for your unknown mixture (and the unknown number) and discuss the accuracy of your results by propagating the errors from the ks. The graphs and tables should be sequentially numbered so that these can be referred to as, say, Figure 1 or Table 1 in your written text. Discuss any possible systematic errors in the experiment and what you could have done to eliminate these.

Questions

1. The percent transmittance of a sample is 24.7% in a cell with a 5.00 cm path length. What is the percent transmittance of the same sample in a cell with a 1.00 cm path length?

2. The absorbance of an 8.54×10^{-5} M $AuBr_4^-$ solution was determined to be 0.410 at 382 nm in a 1.00 cm cell. Assuming that the absorbance is due entirely to $AuBr_4^-$, what is the molar absorptivity of $AuBr_4^-$ at 382 nm?
3. A mixture containing MnO_4^- and $Cr_2O_7^-$ was analyzed spectrophotometrically at 440 and 545 nm. The observed absorbances were 0.385 and 0.653, respectively, at each wavelength for a 1.00 cm cell, and due entirely to MnO_4^- and $Cr_2O_7^-$. The molar absorptivities for MnO_4^- are 93.8 $M^{-1}cm^{-1}$ at 440 nm and 2350 $M^{-1}cm^{-1}$ at 545 nm, while those for $Cr_2O_7^-$ are 370 $M^{-1}cm^{-1}$ at 440 nm and 11 $M^{-1}cm^{-1}$ at 545 nm. Determine the molar concentrations of MnO_4^- and $Cr_2O_7^-$ in the mixture.
4. The molar absorptivity at 500 nm of a compound in aqueous solution is 2500±10 MnO_4^- and $Cr_2O_7^-$. An aqueous solution of the compound in a 3.00±0.01 cm cell shows an absorbance of 0.410±0.005 at 500 nm. What are the molarity and its uncertainty of the compound in the solution?

7: Quantitative Spectrophotometry and Beer's Law (Graph)

Key points:

- Absorbances are additive.
- Higher concentrations have higher absorbance levels.
- For graphs, the closer that R2 is to 1, the better the line.
- Your k-values are equal to the slopes of the line of best fits.

Summarized Procedure:

1. Obtain the absorbances of each diluted solution, the stock solution, and unknown solution.
2. For each ion, graph their absorbances at every dilution for both wavelengths 510 and 575 nm. Add in the line of best fit, and you should have a total of 4 lines.
3. These graphs will be used to determine the k values because the k value is equal to slope of the linear line of the best fit.
4. Knowing the k of each ion at 510 and 575 nm along with the absorbance of the unknowns will allow you to determine the concentration of Co2+ and Cr3+. Refer to the calculations below for an explanation why.

Calculations:

The following system of equations are used to find the concentrations. The highlighted variables are the unknown values you will be solving for. We get the following equations from the fact that absorbances are additive and that the absorbance of a solution is equal to its k value times its concentration (or A = KC). The graphs will provide you with the k values of each ion at each wavelength and the absorbance of the unknown is obtained from your lab data.

$$A_{510} = A_{Cr\ 510} + A_{Co\ 510}$$

$$A_{575} = A_{Cr\ 575} + A_{Co\ 575}$$

$$A_{510} = K_{Cr\ 510}\, C_{Cr} + K_{Co\ 510}\, C_{Co}$$

$$A_{575} = K_{Cr\ 575}\, C_{Cr} + K_{Co\ 575}\, C_{Co}$$

A_{510} = absorbance of unknown at 510 nm

A_{575} = absorbance of unknown at 575 nm

KCr_{510} = k value of chromium at 510 nm

KCr_{575} = k value of chromium at 575 nm

KCo_{510} = k value of cobalt at 510 nm

KCo_{575} = k value of cobalt at 575 nm

CCo = concentration of cobalt

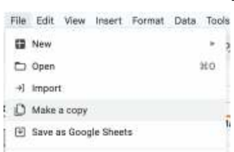
CCr = concentration of chromium

Making the Graphs:

Example Graph	[Lab 7 Computer Beer's Law - Student Edn..xls](#)
Student Graph (MAKE A COPY OF THIS, FILL IT OUT WITH YOUR DATA TO MAKE YOUR GRAPHS)	[Lab 7 Computer Beer's Law - Student Edn..xls](#)

Procedure for Graphs:

1. Click “Student Graph” link above.
2. Click “File” then “Make a Copy.”



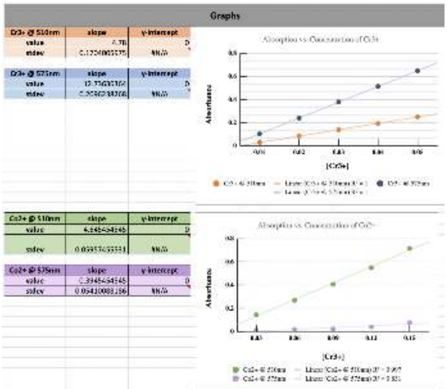
3. Following the instructions on YOUR BLANK COPY, add your experimental data obtained from the lab into the YELLOW HIGHLIGHTED BOXES. The rest of the sheet is set to run the calculations for you. Do NOT adjust the other parts. If you do, you can go back to the original and obtain the necessary functions to fix it.

Data		
Input:		
[Cr3+]	Abs @ 510nm	Abs @ 575nm
0.01	0.029	0.103
0.02	0.084	0.239
0.03	0.138	0.379
0.04	[illegible]	[illegible]
0.05	[illegible]	[illegible]
[Co2+]	Abs @ 510nm	Abs @ 575nm
0.03	0.144	0.012
0.06	0.268	0.018
0.09	[illegible]	[illegible]
0.12	[illegible]	[illegible]
0.15	0.715	0.076
Known Mix	0.413	0.322
Unknown Mix	[illegible]	[illegible]

4. The output part below your data input has data that is used in the calculations. Do not edit this part of the sheet.

Output:	
1	75.60%
2	4800 M-1*cm-1
3	[MnO4-] = 2.73 E-4 M; [Cr2O7-] = 9.73 E-4 M
4	(5.467+/- 0.0725) E-5 M

5. Your graphs will automatically show up in this part of the sheet. The colors of the lines and dots on the graph are coordinated with the results on the left side. On the left, you will find each line’s respective slope value, standard deviation, and y-intercept. The #N/A is correct as there is no standard deviation for the y-intercept. Again, do not edit this part of the sheet.



6. Here are your results for the lab! The k-values correlate with the slopes of each line of best fit. For example, the k-value of Cr^{3+} at 510 nm is equal to the slope of that graph's line of best fit. The standard deviation is also from each slope's respective standard deviation. The data is simply consolidated to this table for your ease. Again, do not edit this part of the sheet.

Results		
Beer's Law Plot	k value (M-1)	StDev (M-1)
Cr3+ @ 510nm	4.78	0.1704005975
Co2+ @ 510nm	4.645454545	0.05957455531
Cr3+ @ 575nm	12.73636364	0.2096238268
Co2+ @ 575nm	0.3945454545	0.05410088186

[Unknown]	
Cr3+	0.02941101495
Co2+	0.06187023158

7. Finally, here are your unknown concentrations. They are calculated for you using the system of equations discussed on page 1. Again, do not edit this part of the sheet.

Results		
Beer's Law Plot	k value (M-1)	StDev (M-1)
Cr3+ @ 510nm	4.78	0.1704005975
Co2+ @ 510nm	4.645454545	0.05957455531
Cr3+ @ 575nm	12.73636364	0.2096238268
Co2+ @ 575nm	0.3945454545	0.05410088186

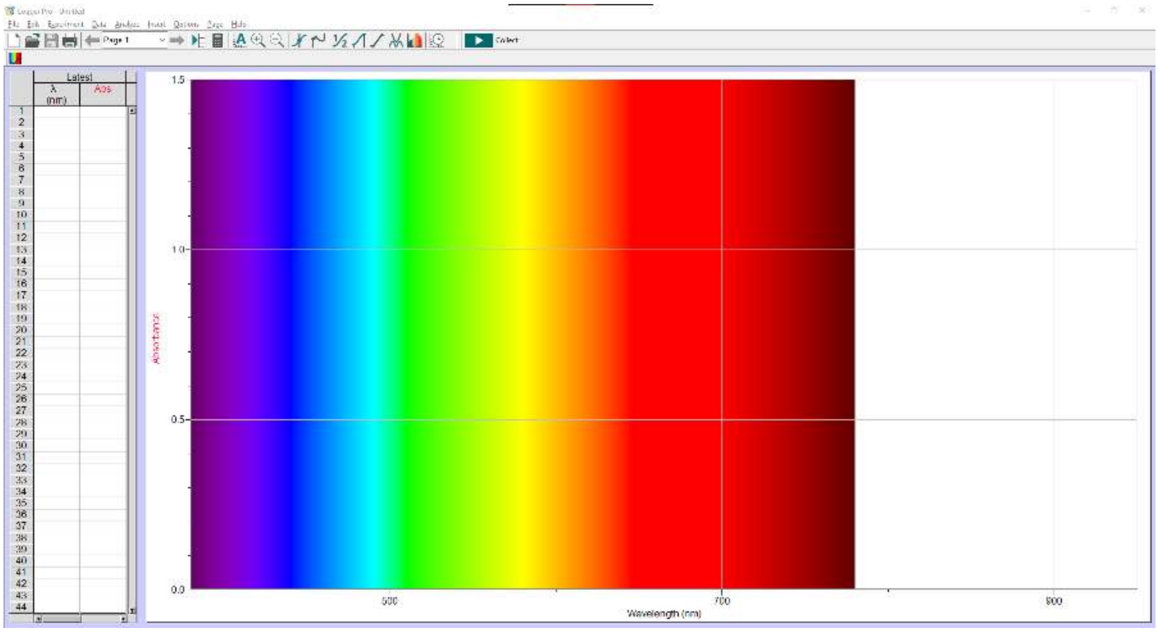
[Unknown]	
Cr3+	0.02941101495
Co2+	0.06187023158

Appendix 3: Instrumental Instructions

Logger Pro with Spectrophotometer

Preparation

1. Use a USB cable to connect the Spectrometer to the computer.
2. Double-click the Logger Pro icon on the desktop.
3. Choose **New** from the **File** menu.



Logger Pro Interface with a Spectrometer Connected (CC BY-NC-SA; Henry Li)

Calibration

1. Fill the cuvette 2/3 full with the blank solution. This amount ensures that all incident radiation passes through the solution. Wipe the outside of the cuvette with Kimwipes. Insert the blank cuvette into the cuvette holder.
2. To calibrate the Spectrometer, go to the **Experiment** menu and choose **Calibrate, Spectrometer: 1**.
3. The calibration dialog box will display the message: "Waiting 90 seconds for lamp to warm up..." After the warm-up is done, the message will change to "Warmup complete." Place the blank cuvette into the cuvette slot of the Spectrometer. Then, choose **Finish Calibration** and wait for the calibration to complete; then choose **OK.** You have finished calibrating your Spectrometer.

Measure the Full Spectrum Absorbance of a Sample (Absorbance vs. Wavelength)

1. Place the standard of interest in the cuvette slot.
2. To measure the full spectrum from 390 nm to 950 nm, select the green **Collect** icon. The absorbance *vs.* wavelength spectrum will be displayed and updated continuously.
3. Select the red **Stop** icon to halt the data collection.
4. Store the spectra by choosing **Experiment**, **Store Latest Run**.
5. The run may be renamed by double-clicking on the column header, by default it is named "Run 1." A dialog box titled "Data Set Options" will pop up. You may change the name and/or place comments about the run in the comment box. Click **OK** to save your changes.
6. You may examine the absorbance at different wavelengths by selecting **Analyze, Examine**, or the keyboard shortcut **Ctrl+E**, and moving the cursor over the spectra. Select **Analyze, Examine** again to exit out of examination mode.

Set up Spectrophotometer for Measuring Absorbance *vs.* Concentration (Beer's Law Experiment)

1. From the full spectra, determine the optimal wavelength for creating this standard curve. (See Experimental Procedure)
2. To set up the data collection mode and select a wavelength for analysis, select the rainbow-colored **Configure Spectrometer** icon to the left of the green **Collect** button.
3. In the pop-up window, change **Collection Mode** to **Absorbance *vs.* Concentration**. Enter Column Name (Concentration), Short Name (Conc.), and Units (mol/L) (they should be entered for you by default). From the drop-down menu, change **Single 10 nm Band** to **Individual Wavelengths**.
4. Click **Clear Selection** in the middle. Locate an absorbance peak of interest on the graph to the right and click on that peak value to select its wavelength. Select **OK** once you are done.
5. Two graphs are now displayed on the screen, a Full spectra graph and a graph of Absorbance *vs*. Concentration with absorbance at the desired wavelength(s).

Conducting the Beer's Law Experiment

1. Place your first Beer's Law standard solution in the spectrometer and select the green **Collect** icon. When the absorbance has stabilized, tap the **Keep** Icon next to the **Collect** icon. Enter the concentration of the first standard in the pop-up window and select **OK**.
2. Repeat the prior step to collect absorbance readings of the remaining set of standards.
3. Choose **Experiment**, **Store Latest Run**.

Display a Graph of Absorbance *vs.* Concentration with a Linear Regression Curve

1. From the **Options** menu, choose **Graph Options**.
2. In the **Axes Options** tab, change **Scaling** to **Autoscale From 0,** and select **Done**.
3. From the **Analyze** menu choose **Curve Fit...**
4. In the pop-up window, under **General Equation:** select **Linear** as the fit equation. The linear-regression statistics for these two data columns are displayed for the equation in the form of

$$y = mx + b$$

   where x is concentration, y is absorbance, m is the slope, and b is the y-intercept. **Note:** One indicator of the quality of your data is the size of b. It is a very small value if the regression line passes through or near the origin. The correlation coefficient, r, indicates how closely the data points match up with (or fit) the regression line. A value of 1.00 indicates a nearly perfect fit.
5. Select **OK**. The graph should indicate a direct relationship between absorbance and concentration, a relationship known as Beer's law. The regression line should closely fit the five data points *and* pass through (or near) the origin of the graph.

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